

CIRCUIT THEORY II - R4053 - 2025	
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T.P. No. 7: Design using classical filtering	
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Submission date: 22/10/2025	Approval date:
Remarks:	

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Exercise #1

Exercise1 Page 1

Sunday, September 28, 2025

11:27

Exercise 1. Within an adapted system, whose characteristic impedance is 50 ohms, it is requested to design a network that allows obtaining an attenuation of 20 d. Simulate the proposed design and verify its operation.

datos: $Z_0 = 50 \Omega \rightarrow$ Impedancia característica

$\alpha = 20 \text{ dB} \rightarrow$ Atenuación from 2046

Dado Que ne qdo feu Coral $\alpha_{\text{atenua}} \Rightarrow$ to now α_{atenua}

Pore α_{atenua} "e duig uene de α_{atenua} "

Impedancia característica — Era

$i = \text{Imagem}$
Simétricas

$$Z_0 = \sqrt{\frac{Z_{i1}}{Z_{i2}}} = dde > Z_{ea}$$

$Z_{i1,2}$

2l diner α_{atenua} lo α_{atenua} matriz [T]:

$$[T] = \begin{bmatrix} \sqrt{\frac{Z_{i1}}{Z_{i2}}} \cosh(\alpha) & \sqrt{Z_{i1} \cdot Z_{i2}} \sinh(\alpha) \\ \sinh(\alpha) & \cosh(\alpha) \end{bmatrix}$$

NS > hee)

dande == $x + 7h$

$X = \text{EN de } \alpha_{\text{atenua}}$
 $p = \text{carr de desplazamiento de } \alpha_{\text{atenua}}$

Eo α_{atenua} : $\alpha = \sqrt{A \cdot D} + \sqrt{B \cdot C}$

$E^E - \text{Caht Eli nenhi } \epsilon$

$\alpha_{\text{atenua}} \rightarrow \alpha_{\text{atenua}} \rightarrow 2A = 2,305\% 2 \text{ M4 el}$
 $\alpha_{\text{atenua}} = 20 \text{ dB}$

$\alpha_{\text{atenua}} \rightarrow \alpha_{\text{atenua}} \Rightarrow \alpha_{\text{atenua}} = 0$

o α_{atenua} : $\alpha = \alpha + 20 = X$

des ant 2 $Z_0 = 50 \Omega \Rightarrow Z_{i1} = Z_{i2} = Z_0 = 50 \Omega$
 $\alpha_{\text{atenua}} = 2,302582$

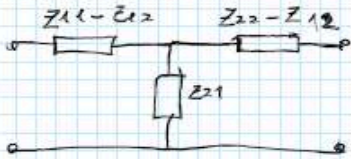
2.6) Calcular la matriz [T]

$\Rightarrow [T] = \begin{bmatrix} \sqrt{\frac{50\Omega}{50\Omega}} \cosh(2,3025) & \sqrt{50\Omega \cdot 50\Omega} \sinh(2,3025) \\ \frac{1}{\sqrt{50\Omega \cdot 50\Omega}} \sinh(2,3025) & \sqrt{\frac{50\Omega}{50\Omega}} \cdot \cosh(2,3025) \end{bmatrix}$

$[T] = \begin{bmatrix} 5,05 = A & 247,4785 = B \\ 0,0099 = C & 5,05 = D \end{bmatrix}$

3.2) Elige un tipo de red canónica y busca los parámetros Z

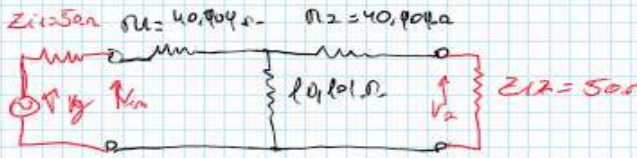
• Circuitos Y



$[Z] = \begin{bmatrix} Z_{11} = \frac{A}{C} & Z_{12} = \frac{A}{C} \\ Z_{21} = \frac{1}{C} & Z_{22} = \frac{D}{C} \end{bmatrix}$

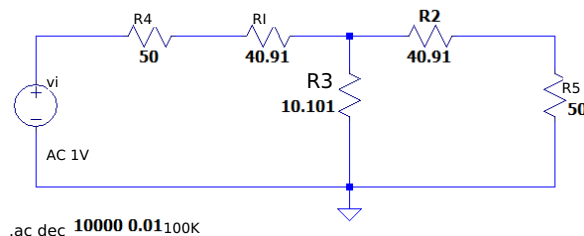
$[Z] = \begin{bmatrix} 51,01 & \approx 10,101 \\ 10,101 & 51,01 \end{bmatrix}$

$Z_{12} = 50\Omega$ $R_1 = 40,904\Omega$ $R_2 = 40,904\Omega$



• Solucion la red em negro) obtenes o 20 dB, The
 agregan algunos dede atenuación para que sea adoptados

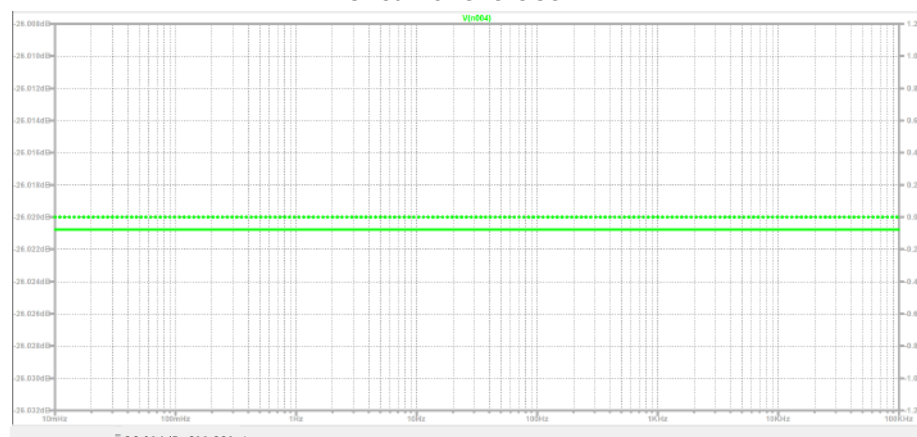
Simulation:



.ac dec 10000 0.01 100K

Graph 1

Circuit of exercise 1.



Graph 2

Frequency response of the voltage transfer of the circuit from exercise 1.

Exercise #2

Given a PI-type high-pass network, obtain the image parameters.

2) Parámetros imagen π resaca Ho

System simétrico

Diagram: A π -network with series admittance Y_1 and shunt admittances Y_2 and $Y_2/2$ at the input and output respectively. The input impedance is Z_1 and the output impedance is Z_2 .

$\rightarrow Y_1 = \frac{1}{sL}$
 $\rightarrow Y_2 = \frac{1}{sL}$
 $\rightarrow Y_{0\pi} = \sqrt{\Delta Y}$

$Y = \begin{bmatrix} Y_A + Y_B & -Y_B \\ -Y_B & Y_B + Y_A \end{bmatrix} = \begin{bmatrix} L + Y_1 & -Y_1 \\ -Y_1 & \frac{Y_2}{2} + Y_1 \end{bmatrix}$

$\Delta Y = Y_{11} \cdot Y_{22} - Y_{21} \cdot Y_{12} = \left(\frac{Y_2}{2} + Y_1\right)^2 - (-Y_1)^2$
 $\Delta Y = \frac{Y_2^2}{4} + \frac{Y_2}{2} \cdot Y_1 \cdot 2 + Y_1^2 - Y_1^2 = \frac{Y_2^2}{4} + Y_2 Y_1$

$Y_{0\pi} = \sqrt{\frac{Y_2^2}{4} + Y_2 Y_1} = \sqrt{Y_2 Y_1 \left(\frac{Y_2}{4 Y_1} + 1\right)} = \sqrt{Y_2 Y_1 \left(\frac{Y_2}{4 Y_1} + 1\right)} = \sqrt{Y_2 Y_1} \sqrt{\frac{Y_2}{4 Y_1} + 1} = \sqrt{\frac{C}{L}} \sqrt{\frac{1}{4 s^2 C L} + 1}$

$Z_{0\pi} = \frac{1}{Y_{0\pi}} = \frac{1}{\sqrt{Y_2 Y_1 \left(\frac{Y_2}{4 Y_1} + 1\right)}} = \sqrt{\frac{L}{C}} \sqrt{\frac{4 s^2 C L}{1 + 4 s^2 C L}}$

$\rightarrow Z_{s1} = \sqrt{\frac{AB}{CD}} = \sqrt{\frac{Z_{11}}{Y_{11}}} = \sqrt{\frac{\frac{Y_2 + Y_1}{Y_1}}{\frac{Y_2}{2} + Y_1}} = \sqrt{\frac{\frac{Y_2 + Y_1}{Y_1}}{\frac{Y_2}{2} + Y_1}} = \sqrt{\frac{1}{\frac{Y_2^2}{4} + Y_2 Y_1}} = Z_{0\pi}$

$\rightarrow Z_{s2} = Z_{s1} = Z_{0\pi}$

Exercise #3

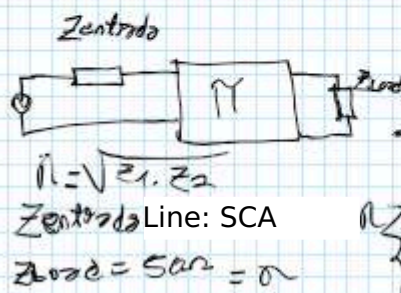
Exercise3 Page 1

Tuesday, September 30, 2075

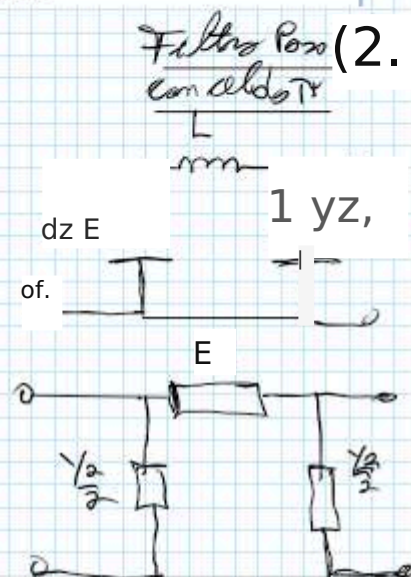
10:10

Exercise 3. Design a low-pass filter with the following specifications:

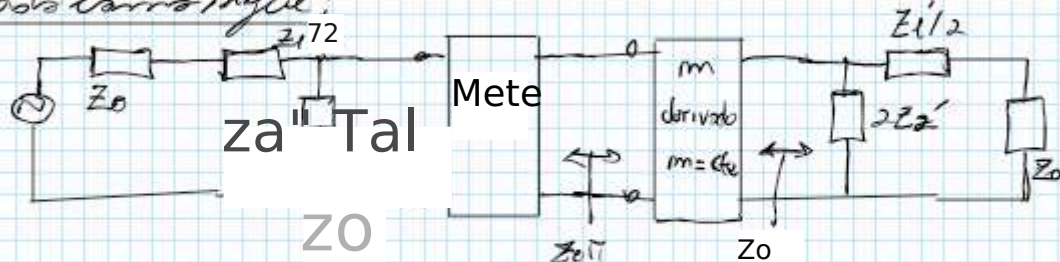
- Must present adaptation to source and load, both of Z_0 ohms
- minimize the use of inductors, PI prototype cells will be used
- The prototype cell must present infinite impedance at a frequency of 1 GHz
- The filter must present infinite attenuation at 1.1GHz



- Z_{load} en $f_c = 1.64 \text{ GHz}$
- $\omega \rightarrow \infty$ en $f_{cut} = 1.1 \text{ GHz}$



• Para un diseño de filtros, utilizamos un "filtro compuesto" de dos ceros sigue:



• Este esquema nos permite adaptar tanto a la entrada como la salida al terminador de la carga y generador.

e A la que "m" denota permitividad relativa ϵ_r y t_n espesor de la capa.

TO gora 7 $Moe = 125 \text{ } \mu\text{E Ji}$

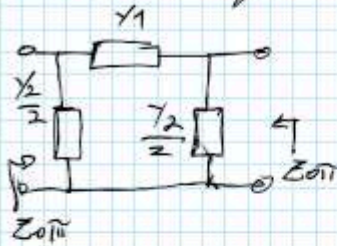
Mun inma oe 125 JC ha.

$$m = 0.6$$

Exercise! Sheet_2

Tuesday, 9 of septiembre of 2025 18:06

1.º Comenzamos trabajando el bloque Kcto



$$\omega_c = \frac{2}{\sqrt{LC}}, \quad f_c = \frac{1}{\pi \sqrt{LC}}$$

$$\text{De los bloques} \Rightarrow Y_1 = \frac{1}{3\omega L}$$

$$Y = 3\omega C$$

$$Z_0 = 50 \Omega = \sqrt{\frac{L}{C}} \Rightarrow L = 15192 \text{ nH}$$

$$1.6 \text{ GHz} = \frac{1}{\pi \sqrt{LC}} \Rightarrow C = 6136 \text{ pF}$$

$$C = 3189 \text{ F}$$

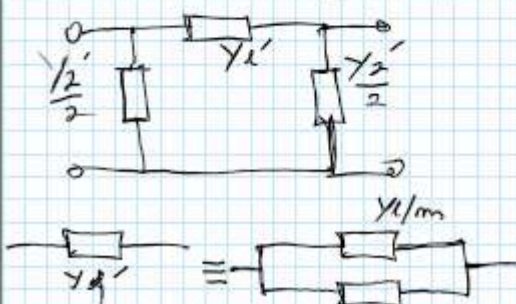
2.º Trabajamos con el bloque "m-densado"

«Do enunciado» $\Rightarrow$ atenuación ∞ en $f_{\infty} = f_c$ or

$$\Rightarrow 1.16 \text{ GHz} = 1.6 \text{ GHz} \cdot \frac{1}{\sqrt{1-m^2}} \Rightarrow m = 0.42$$

• Dicho atenuación ∞ en f_{∞} se debe a la celda Kcto

• celda π tipo m-densado



$$Y_1' = \frac{1}{m} + \frac{Y_2}{4m} (1-m^2)$$

$$Y_2' = m \cdot Y_2$$

$\Rightarrow$ donde Y_1 y Y_2 son en el celda π de 1.º

cuentos

$$Y_1' = \frac{1}{m} + \frac{Y_2}{4m} (1-m^2) \Rightarrow L' = L \cdot m$$

$$\text{bo } 15192 \text{ nH} \cdot 0.42 =$$

$$L' = 6388.4 \text{ nH}$$

$$Y_2' (1-m^2) = 3\omega C \cdot \frac{(1-m^2)}{4m}$$

$$C' = C \cdot \frac{(1-m^2)}{4m} = 6136 \text{ pF} \cdot 0.49$$

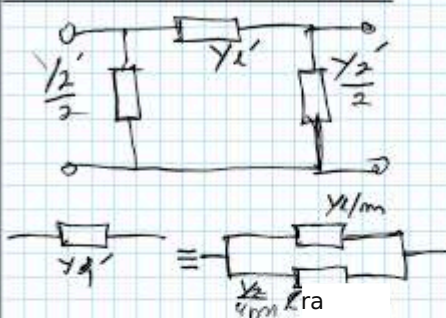
$$C' = 3026.44 \text{ pF}$$

$$m \cdot Y_2 = m \cdot 3\omega C \Rightarrow C \leq m \cdot C$$

$$C_2 = 2169 \text{ pF} \quad C_2' = 1333.6 \text{ pF}$$

UB Trabajo con It hemisecciones the antena y el solido

- vortade una celsoo tipo II y luego of los Pontelett
- celsoo π tipo



$$Y_1' = \frac{Y_1}{m} + \frac{Y_2}{4m} (1-m^2)$$

$$Y_2' = m \cdot Y_2$$

Donde Y_1 y Y_2 Tienen de late π enigich de Kete voligado en LE

Cuentos

$$Z = \frac{l}{5w} \frac{l}{m} \Rightarrow L' = 1.1m =$$

$$L' = 15.42mH \cdot 0.6 = 9.552mH = L'$$

$$\frac{Y_2}{4m} (1-m^2) = 5w \frac{(1-m^2)}{4m}$$

$$C_1' = C \cdot \left(\frac{1-m^2}{4m} \right) = 6.36PF \cdot \frac{(1-0.6^2)}{4 \cdot 0.6}$$

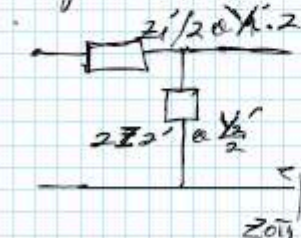
$$C_1' = 1.62PF$$

$$0.6 \cdot Y_2 = m \cdot 5w \Rightarrow C_2' = m \cdot C = 3.816PF$$

$$\frac{C_2'}{2} = 1.908PF$$

• Nota: Deoseme $m = 0.6$ cma antena of disena. Puste que permite

- En forma de los datos obtenidos $\Rightarrow$ planteo la hemisección



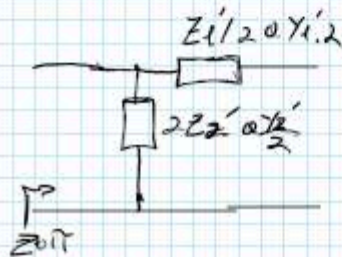
$$2Y_1' = 2 \left(\frac{Y_1}{m} + \frac{Y_2}{4m} (1-m^2) \right) = 2 \left(\frac{1}{5w \cdot m} + 5w \frac{(1-m^2)}{4m} \right)$$

$$2Y_1' = \frac{1}{5w \cdot \frac{L}{2}} \text{ ra } 5w \cdot 2 \cdot C \cdot \frac{(1-m^2)}{4m}$$

$$\Rightarrow L' = \frac{L \cdot m}{2}, C_1' = 2 \cdot C \cdot \left(\frac{1-m^2}{4m} \right)$$

$$\Rightarrow L' = \frac{15.42mH \cdot 0.6}{2}, C_1' = 2 \cdot 6.36PF \cdot \frac{(1-0.6^2)}{4 \cdot 0.6}$$

$$L' = 4.177mH, C_1' = 3.384PF$$



$$\frac{Y_2'}{2} = \frac{m \cdot Y_2}{2} = \frac{m}{2} \cdot 5w \Rightarrow C_2' = \frac{C \cdot m}{2} = 1.908PF$$

$$f_{ao} = \frac{fc}{\sqrt{1-m^2}} = \frac{1848}{\sqrt{1-0.6^2}} = 1.25642$$

Wednesday, October 1, 2025 11:44



Circuit of exercise 3.



Frequency response of the voltage transfer of the circuit from exercise 3.

Exercise #4

Exercise4 Sheet 1

Wednesday, October 1, 2025

11:57

Exercise 4. With the objective of eliminating frequencies, espurias, an E-constant filter is placed in a 50-ohm system, according to the following schematic:



Se asks:

1. Obtain for what frequency value within the passband is the filter matched.
2. Obtain the attenuation of the filter for said frequency

and item to point i, if $R_g=R_l$ is changed to a resistor of 1 ohms

4. Obtain the value of $R_g=EL$ so that the system is matched to a frequency of 100 kHz
5. Simulate the proposed design and verify its operation.

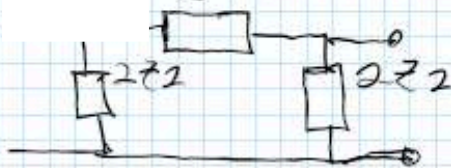
MA

el diagrama

Le

Mi

1 fp (to Leo 4



$$Z_{ay} = PNL$$

$$\frac{1}{\sqrt{1 + \dots}}$$

A Zool?

we frequency de

$$we = \frac{1}{2\sqrt{LC}}$$

$Z_{in} \rightarrow \infty$

Item 1)

Elta starts by the $Z_{in} = 50 \Omega = R_g = R_L$

Sd UE

$$Z_{in} = R$$

$$R = \sqrt{\frac{Z_1 \cdot Z_2}{n^2}}$$

ZA 1,

$$2Z_2 = H_0, La L_1 = L_2 = 2 \mu H$$

$$R = \sqrt{\frac{1}{\omega^2 C} \cdot \frac{1}{\omega^2 C}} = \sqrt{\frac{1}{\omega^2 C^2}}$$

$$\text{or } Z_{IT} = \sqrt{\frac{1}{\omega^2 C^2}} \cdot \frac{1}{\sqrt{1 + \frac{1/\omega^2 C}{2\omega L}}} > \sqrt{\frac{1}{\omega^2 C^2}} \cdot \frac{La}{\sqrt{1 + \frac{1}{2\omega^2 L^2 C}}}$$

$$\frac{1}{50 \Omega} = \sqrt{\frac{1}{\omega^2 C^2}} \cdot \frac{1}{\sqrt{1 - \frac{1}{2\omega^2 L^2 C}}} \Rightarrow \sqrt{1 - \frac{1}{2\omega^2 L^2 C}} = \frac{\sqrt{1}}{R_g}$$

$$\sqrt{1 - \frac{1}{2\omega^2 L^2 C}} = \frac{1}{5}$$

$$\left(\sqrt{1 - \frac{1}{25}}\right)^2 = \left(\frac{1}{5}\right)^2 \Rightarrow 1 - \frac{1}{25} = \frac{1}{25} \Rightarrow 1 - \frac{1}{25} = \frac{1}{25}$$

$$\Rightarrow \omega^2 = \frac{1}{2CL} \cdot \frac{1}{1 - \frac{1}{25}} \Rightarrow \omega = \sqrt{\frac{1}{2CL} \cdot \frac{1}{1 - \frac{1}{25}}} =$$

$$\cdot 1 - \frac{1}{25} \text{ solo de } \left(\frac{\omega}{\omega_0}\right)^2 \text{ con and } \sqrt{\frac{1}{2} \cdot 10} \cdot 10 = 5 \text{ cm}$$

• conclusion: $\omega_0 = 511 \text{ M rad/s}$ en la banda de paso para filtros
de paso, ω_0 de paso, ω_0 de paso donde $Z_{in} = R_g = R_L$
• $f = \frac{\omega}{2\pi} = 81.2 \text{ kHz}$, $\omega_0 = \frac{1}{2\sqrt{LC}} = \text{SM}$ $H \Rightarrow$ usar sweep stake

Item 2)

de ρ de $\alpha = \text{atenuación}$ para $\omega = 511 \text{ M rad/s}$, $f = 81.2 \text{ kHz}$
De la función of propagación $\rightarrow e^{\alpha} = e^{\alpha + j\beta}$

$$\cdot e^{\alpha} = \frac{V_1}{V_2} \Big|_{Z_{in}} = \sqrt{A \cdot D} + \sqrt{B \cdot C}$$

$$\cdot e^{\alpha} = A + \sqrt{B \cdot C} \Rightarrow \begin{cases} A = -\frac{Y_{22}}{Y_{21}} \\ B = -\frac{1}{Y_{21}} \\ C = \frac{\Delta Y}{-Y_{21}} \end{cases}$$

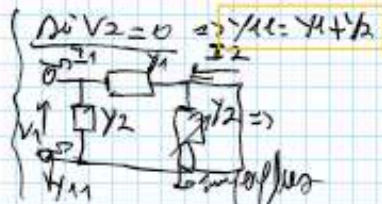
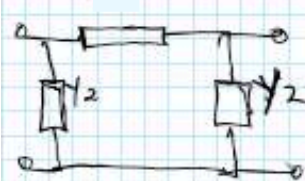
And

$$\begin{aligned} I_1 &= Y_{11} \cdot V_1 + Y_{12} \cdot V_2 \\ I_2 &= Y_{21} \cdot V_1 + Y_{22} \cdot V_2 \end{aligned}$$

$$\Rightarrow \begin{cases} Y_{11} = \frac{I_1}{V_1} \Big|_{V_2=0} \\ Y_{21} = \frac{I_2}{V_1} \Big|_{V_2=0} \end{cases}$$

$$Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0}$$

$$Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0}$$



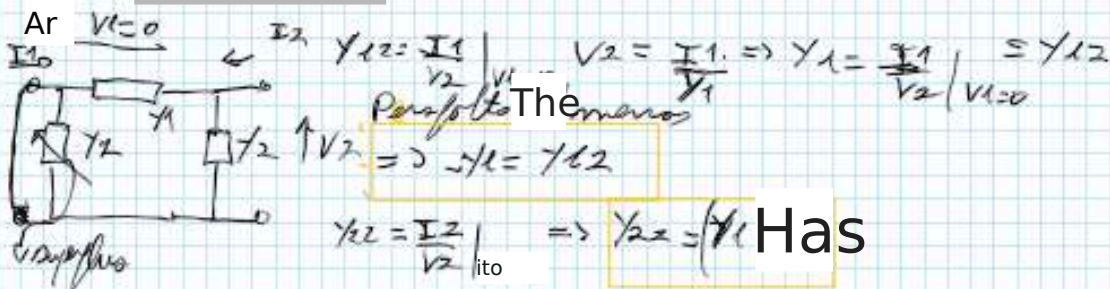
$$\text{VAR } -\frac{I_2}{V_1} \Rightarrow Y_{12} = \frac{I_2}{V_1} \Big|_{V_2=0} \Rightarrow Y_{12} = y_{aa}$$

Por simetría $Y_{12} = Y_{21}$

Exercise4 Sheet_3

Sunday, of October of

12-21



• Encontrando Y_{11}, Y_{12}, Y_{21} y Y_{22} :

$$Y_{12} = Y_{21} = -g_c$$

$$Y_{11} = Y_{22} = Y_1 + Y_2 = \text{of } \frac{1}{g_L} = \frac{g_L^2 C_L + 1}{g_L} = Y_{11} = Y_{22}$$

$$\Delta Y = \begin{vmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{vmatrix} = Y_{11} Y_{22} - Y_{12} Y_{21} = \left(\frac{g_L^2 C_L + 1}{g_L} \right)^2 - (-g_c)^2$$

$$E \quad \frac{g_L^2 C_L + 1}{g_L} E \quad g_c^2 C^2 = \frac{(g_L^2 C_L + 1)^2}{g_L^2} - g_c^2 C^2$$

$$(g_L^2 C_L + 1)^2 = g_L^4 C_L^2 + 2g_L^2 C_L + 1$$

$$\Delta Y = \frac{g_L^4 C_L^2 + 2g_L^2 C_L + 1}{g_L^2} - g_c^2 C^2 = \frac{2g_L^2 C_L + 1}{g_L^2} = \Delta Y$$

• Buscamos los parámetros A, B, C

$$A = \frac{-Y_{22}}{Y_{21}} = \frac{g_L^2 C_L + 1}{g_L} = \frac{g_L^2 C_L + 1}{g_L^2 C_L} = A$$

$$B = -\frac{1}{Y_{21}} = \frac{1}{g_c} = \frac{1}{g_c} = B$$

$$C = \frac{\Delta Y}{-Y_{21}} = \frac{2g_L^2 C_L + 1}{g_L^2} = \frac{2g_L^2 C_L + 1}{g_L^2} = C$$

$$B \cdot C = \frac{1}{g_c} \cdot \frac{2g_L^2 C_L + 1}{g_L^2} = \frac{2g_L^2 C_L + 1}{g_c g_L^2}, \quad \sqrt{B \cdot C} = \frac{\sqrt{2g_L^2 C_L + 1}}{g_c g_L}$$

Exercise4 Hoja_4

Sunday, 10 October 2025 12:48

$$CE \stackrel{\sim 0}{=} A + \sqrt{BC} = \frac{s^2 CL + 1}{s^2 CL} \text{ Do you see}$$

$$e^{\alpha} \cdot e^{j\omega} = \frac{s^2 CL + 1 + \sqrt{2s^2 CL + 1}}{s^2 CL} \quad (s = j\omega)$$

$$e^{\alpha} e^{j\omega} = e^{\frac{-W^2 CL + 1 + \sqrt{-2W^2 CL + 1}}{-W^2 CL}} \quad , \text{ con } Wc = \frac{1}{\sqrt{LC}} =$$

$$\text{Si } W = 5.1 \text{ rad/s} \Rightarrow \begin{cases} -W^2 CL + 1 = e^{-0.799} \quad , \quad 1 \\ -W^2 CL = -0.5202 \quad \text{and } -2.6010^{-15} \\ \sqrt{-2W^2 CL + 1} = 0.25 \end{cases}$$

$$\alpha e^{j\omega} = \frac{0.4799}{-0.5202} + \frac{0.25}{-0.5202} = \frac{-0.9223}{2} - \frac{0.3845}{5}$$

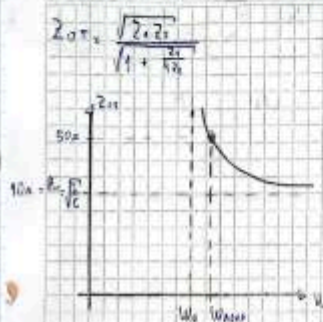
$$\alpha = |e^{\alpha}| = \sqrt{a^2 + b^2} = 0.9992 = E \Rightarrow \ln(0.9992) = \alpha \ln(e)$$

bred paraminimo con
pra introduce formula

$$-8.0032 \cdot 10^{-4} = \alpha \cdot 1$$

or $\approx \alpha$

Justificación teórica: Dado que el filtro se encuentra en la salida de la etapa de entrada, el filtro se encuentra en la salida de la etapa de entrada. En la salida de la etapa de entrada, el filtro se encuentra en la salida de la etapa de entrada. En la salida de la etapa de entrada, el filtro se encuentra en la salida de la etapa de entrada.



$$Z_{in} = \frac{1}{sC} \parallel Z_{in} = \frac{1}{sC} \parallel \frac{1}{sC} = \frac{1}{sC} \parallel \frac{1}{sC}$$

$$Z_{in}(W) = \frac{10 \Omega}{\sqrt{1 - \frac{1}{W^2 L^2 C^2}}} = \frac{10 \Omega}{\sqrt{1 - \left(\frac{W_c}{W}\right)^2}}$$

En la salida de la etapa de entrada, el filtro se encuentra en la salida de la etapa de entrada. En la salida de la etapa de entrada, el filtro se encuentra en la salida de la etapa de entrada.

$$W_c = 5.1 \text{ rad/s}$$

$$W_c^2 = \frac{1}{LC} \Rightarrow W = 5.1 \text{ rad/s}$$

Exercise4_Hoja_5

Sunday, of October 2025 14:09

Item 3) Evolucionar

It

expansion de Z_{out} para adaptacion a LAn

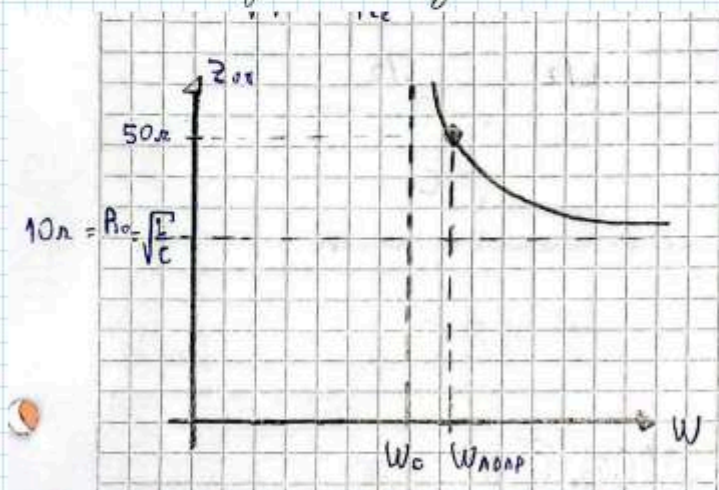
$Z_{in} = \frac{1}{\frac{1}{Z_0} + \frac{1}{Z_L}}$ **You** $\Rightarrow \sqrt{1 - \frac{L}{2WC L}} = \sqrt{\frac{1}{2} \cdot \frac{1}{2}}$

a- $\frac{1}{2WC L} = \frac{1}{2} \cdot \frac{1}{2} \Rightarrow 1 - \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2WC L} \Rightarrow (1 - \frac{1}{2} \cdot \frac{1}{2}) 2WC L = 1$

$\Rightarrow (2CL - \frac{1}{2} \cdot 2CL) \omega^2 = 1 \Rightarrow (2CL - L^2) \omega^2 = 1 \Rightarrow \omega = \sqrt{\frac{1}{2CL - L^2}}$

Reemplazando $\Rightarrow \omega = \sqrt{\frac{1}{2 \cdot 10^{-8} \cdot 2 \cdot 10^{-4} - 2 \cdot 10^{-4}^2}} = 501.996,025$

donde $\omega \neq \omega_{res}$ $\Rightarrow$ Alumno, se concluye that no hay una ω tal que se pueda adaptar el circuito hacia $1 \Omega = 0 \Omega = 0 \Omega$ puesto que ω debe ser un valor real y no un imaginario



Sustitución Teórica: Dado que como se puede observar en el gráfico, Z_{out} solo puede tener valores reales positivos o negativos en el rango de los ω $\rightarrow$ no se $\Rightarrow$ existen frecuencias que cumplen LAn the

Exercise4 oja_6

Sunday, 5 of October 2025 17:16

Item 4)

Buscamos $R_g = R_L = Z_{0T}$ tal que $f = 100 \text{ kHz} \Rightarrow \omega = 2\pi f = 200\pi \text{ rad/s}$

$$Z_{0T} = R_g = R_L = \text{THE } \frac{1}{\sqrt{1 - \frac{1}{2\omega^2 CL}}} = \sqrt{\frac{Z_{0T}^2 \cdot \frac{1}{2}}{2\omega^2 CL}}$$

$$Z_{0T} = R_g = R_L = 10 \cdot \frac{1}{\sqrt{1 - 63,32574}} = -1,2665$$

From $Z_{0T} = -1,2665$

Impedancia de capacitancia
nóla para adaptación a $f = 100 \text{ kHz}$

$$Z_{0T} = Z_C = \frac{1}{\omega C} = +1,2665$$

$$\frac{1}{\omega \cdot 1,2665} = C$$

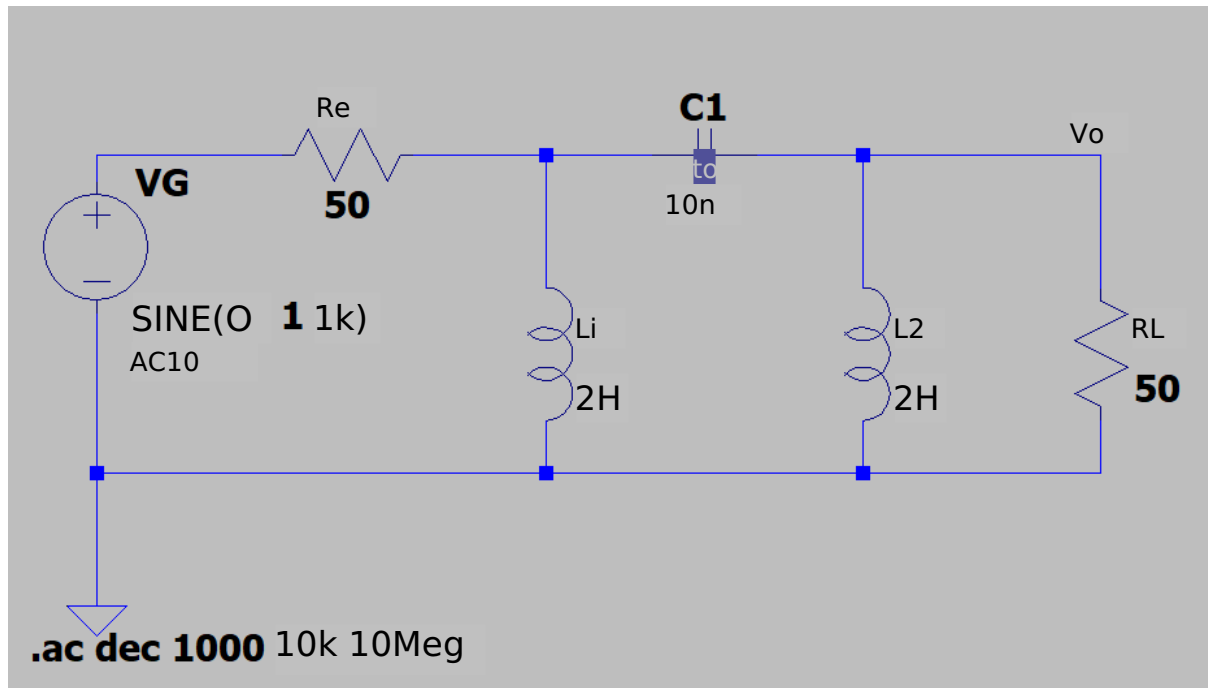
$$\frac{1}{2\pi \cdot 100 \text{ kHz} \cdot 1,2665} = C \approx 1,25 \mu\text{F}$$

Note: Se lo puede justificar
con un átomo, o lo con guías

Item 5)

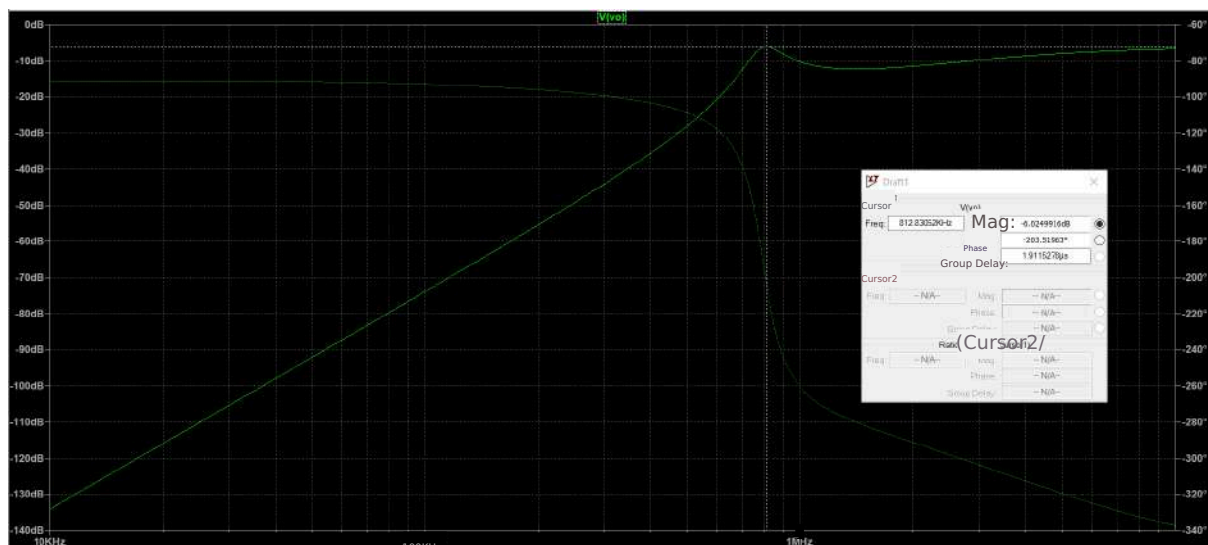
Simulación: En la simulación, hay una caída de -6 of
para la frecuencia deseada de $f = 812 \text{ Hz}$, $R_g = 5 \Omega$, $R_L = 5 \Omega$
es " - 6 dB of por la introducción de carga input/output
a la red PI. Puesto que, la red no introduce a 20

Simulation:



Graph 5

Circuit of exercise 4.



Graph 6

Frequency response of the voltage transfer function of the circuit from exercise 4.

Exercise #5

Exercises Sheet 1

Sunday, October 5, 2025 17:34

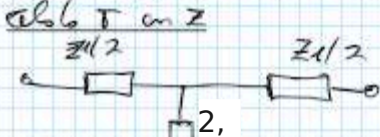
Exercise 5, Build a filter with a cutoff frequency in 1MHz that presents in 1,2MHz uma attenuation of 20dB. Calculate the number of cells required to meet the specification assuming the cells are constant-E filters. Simulate the proposed design and verify its funcionamiento.

Date:-

$f_c = 1 \text{ MHz}$, $\alpha = 20 \text{ dB}$ and $f = 1.2 \text{ MHz}$ 1. Also "Klee"

funr— that ^{tenue} en FF ; 2d fe ^{esistente. e} ^{impresion}
 a veros que tenemos en futoros los p.

1 The Extalbar, interna, Lisena → slo I. condens
con +1 alle fo, baler santo oteno Aihō delo and IA
aronto cero del mismo tifo hoz qen ... con br. v. m.



$$Z_{OT} = R \cdot \sqrt{1 + \frac{Z_1}{4Z_2}}, \quad R = \sqrt{Z_1 \cdot Z_2}$$

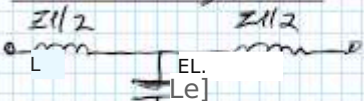
(4720 ENE.

$$Z_{OT} = 0 \quad Z_1 = R_1 \quad Z_2 = \frac{L}{s}$$

$$Z_{01} = \sqrt{E_{VEL} - E_{VE} \sqrt{1 - \frac{u^2}{c^2}}}$$

ZOT=NE AC Y O with E $25=5er$ $2=04=70$

- Abwasser ALG NG F ng molen 50 m



Foo $n\sqrt{1+t}$ E, $\Rightarrow \left(\frac{e}{n}\right)^2 (1/)$

$Z_{0202} = 11710544$

2/13

$$\delta = \delta w \Rightarrow 0 = T - W L_c \Rightarrow W L_c = 1$$

$$W_{\text{eff}} = \sqrt{\frac{4}{L_e}} - 20\%$$

For and The

$$n = \sqrt{\frac{3L + 1}{3L}} = \text{And } \frac{1}{n} = \frac{1}{\sqrt{\frac{3L + 1}{3L}}} = \frac{\sqrt{3L}}{\sqrt{3L + 1}}$$

$$i207 = \sqrt{\frac{1}{2}} \sin \omega = 0 \quad (2)$$

$$\left\{ \begin{array}{l} \text{Wez} \Rightarrow \frac{V_L}{\sqrt{L C}} \quad E) \Rightarrow EE \quad = \frac{2}{\omega C} \Rightarrow L = \left(\frac{2}{\omega C} \right)^2 \cdot \frac{1}{2} \\ Z_{0r} = \sqrt{\frac{L}{C}} \Big|_{\omega=0} = \tan(\alpha) \Rightarrow L = 50^2 \cdot C \\ L = 459 \text{ nH} \end{array} \right.$$

$$\Rightarrow 50^2 \cdot C = \left(\frac{2}{4 \times 10^3}\right)^2 \cdot 1 \text{ rad}$$

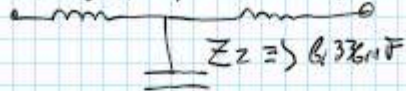
> ES 6366m7

Ejercicio5_Hoja_2

Sunday, 10 October 2025 18:30

• 600 T mas quedaria como:

$$Z_{12} = 7.9595 \mu H \quad Z_{12} = 7.9575 \mu H$$



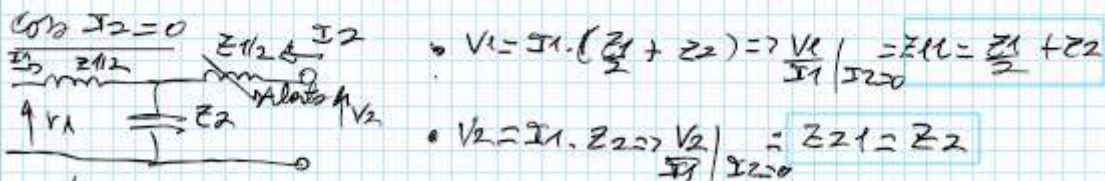
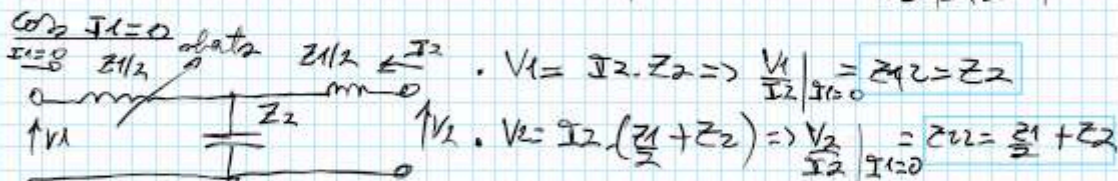
2.9) Planteo C^0, A, D, C, D, C^T

• Por otro lado $C^0 = \sqrt{A D} + \sqrt{D C}$ y para encontrar $C^0 \Rightarrow A, D$

$$C^0 = C^0 C^0 = A + \sqrt{D C} \quad \left\{ \begin{array}{l} C^0 = C^0 \\ A = \frac{Z_{21}}{Z_{11}}, \quad D = \frac{\Delta Z}{Z_{11}}, \quad E = L \\ C^0 \end{array} \right.$$

• From equations parametric Z .

$$\begin{aligned} V_1 &= Z_{11} I_1 + Z_{12} I_2 \\ V_2 &= Z_{21} I_1 + Z_{22} I_2 \end{aligned} \Rightarrow \left\{ \begin{array}{l} Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0}, \quad Z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0} \\ Z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0}, \quad Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0} \end{array} \right.$$



Matrix Z

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{Z_1}{2} + Z_2 & Z_2 \\ Z_2 & \frac{Z_1}{2} + Z_2 \end{bmatrix} = \begin{bmatrix} \frac{5L}{2} + \frac{1}{5C} & \frac{1}{5C} \\ \frac{1}{5C} & \frac{6L}{2} + \frac{1}{5C} \end{bmatrix}$$

$$[Z] = \begin{bmatrix} \frac{5L}{2} + \frac{1}{5C} & \frac{1}{5C} \\ \frac{1}{5C} & \frac{6L}{2} + \frac{1}{5C} \end{bmatrix}$$

Exercises Hoja_3

Sunday, 5 of October of 2025 18:51

32) ~~Obtenga~~ A, B, C por planteo

$A = \frac{EM}{Z_{21}}, B = \frac{From}{Z_{21}}, C = \frac{l}{Z_{21}}$

$A = \frac{\frac{\$^2 L C + 2}{2 \$ C}}{2} = \frac{\text{aunt}}{2} = A$

Is,

$\Delta Z = Z_{11} \cdot Z_{22} - Z_{12} \cdot Z_{21} = \left(\frac{\$^2 L C + 2}{2 \$ C} \right)^2 - \left(\frac{l}{\$ C} \right)^2 = \frac{\$^4 L^2 C^2 + 4 \$^2 L C + 4}{4 \$^2 C^2} - \frac{1}{\$^2 C^2}$

$\Delta Z = \frac{(\$^4 L^2 C^2 + 4 \$^2 L C + 4)}{4 \$^2 C^2} - \frac{1}{\$^2 C^2} = \frac{\$^4 L^2 C^2 (1/4) + \$^2 L C + 4}{4 \$^2 C^2}$

$\Delta Z = \frac{\$^2 C^2 \left[\$^2 L^2 (1/4) + 1 \cdot \frac{L}{C} \right]}{4 \$^2 C^2} = \frac{\$^2 L^2}{4} + \frac{1}{\$} = \Delta Z$

$B = \frac{\Delta Z}{Z_{21}} = \frac{\frac{\$^2 L^2}{4} + \frac{1}{\$}}{\frac{\$^2 L^2 C + 2}{2 \$ C}} = \frac{\$^3 L^2 C + \$ L}{4} = B$

$C = \frac{1}{\$ C} = \$ C = C$

$\theta = \frac{\$^2 L C + 2}{2} + \sqrt{\left(\frac{\$^3 L^2 C + \$ L}{4} \right) \cdot \$} = \frac{\$^2 L C + 2}{2} + \sqrt{\frac{\$^4 L^2 C^2 + \$^2 L C}{4}}$

$\theta = \text{Bis e E} \quad \frac{-W^2 L C + 2}{2} + \sqrt{\frac{W^2 L^2 C^2 - W^2 L C}{4}}$

• Pero $W = 2\pi \cdot 1.2 \text{ MHz} = 2.4 \cdot 10^6 \text{ rad/s}$

• $\frac{-W^2 L C + 2}{2} = -1.865 \quad \omega L = 15.91 \text{ mH}$

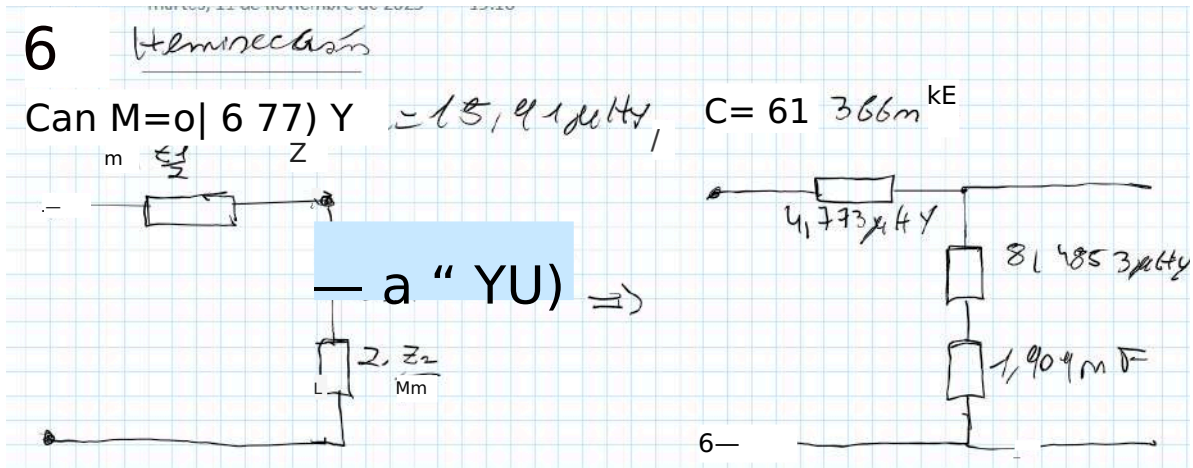
$\Rightarrow e \cdot -1.865 + 1.57 = -0.295$

$\sqrt{\frac{W^2 L^2 C^2 - W^2 L C}{4}} = 1.57 \quad |e^\theta| = 0.295 \Rightarrow \ln(0.295) = \alpha + j\beta$

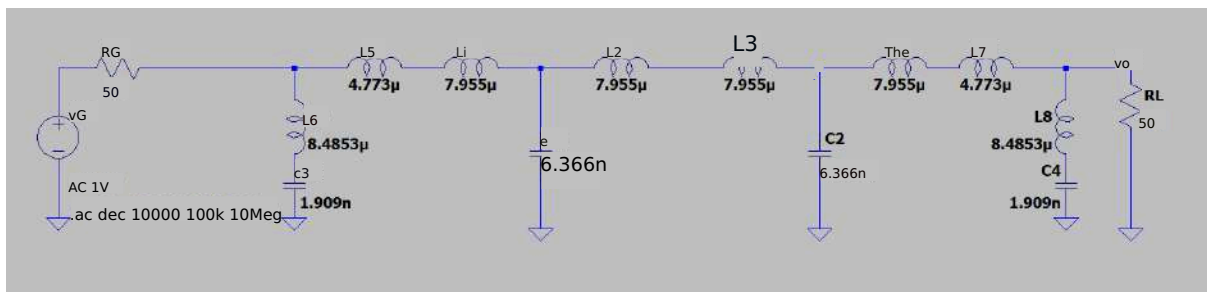
$\omega L = 15.91 \text{ mH}$

A, $22 = \alpha$ (neper)

conclusion: con 2 bobinas de 15.91 mH y 200 nF que tengo a 1.2 MHz se me da $\alpha = -0.54 \text{ dB}$



Simulation:



Graph 7

Circuit of exercise 5.

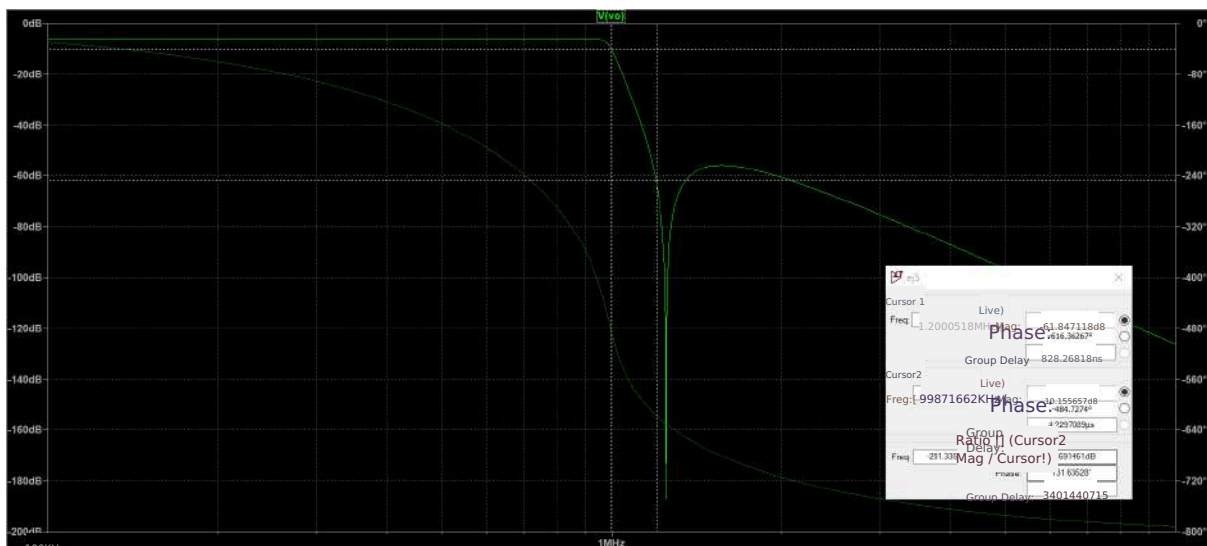


Figure 8

Frequency response of the voltage transfer function of the circuit from exercise 5.

Exercise #6

Obtain a filter with a cutoff frequency of 1MHz that presents a 20dB attenuation at 1.2MHz. Calculate the number of cells needed to meet the specification assuming the cells are m-derivative filters.

Exercises Sheet 1
Friday, October 24, 2025 18:09

Exercise 6. Obtain a filter with a cutoff frequency at 20dB attenuation. Calculate the number of cells needed that presents at 1.2MHz? one to meet the specification assuming the cells are m-derivative filters. Simulate the proposed design and verify its operation.

Datos
 $f_c = 1 \text{ MHz}$, $\alpha = 20 \text{ dB}$ en $f = 1.2 \text{ MHz}$ 1 Choo Wee"

queremos que atenué en $f > f_c$, unos 20dB que es bastante. Por simplicidad de celulas que tenemos un filtro de los tipos.

49) Establezca entenas. The Miner Cat. $\rightarrow$ celula Y. Convergencia
 con 1 dl forum holus iteruo I must and
 avonito celulas del mismo tipo hoy que en un don en. Por cada.

celula T en Z
 $Z_1/2$ $Z_1/2$ TP
 4 2
 $r = 50 \Omega$ $S_{01} = Z_0$
 celula T por cada mE
 zz TT ame o --- 1ef ---

$Z_{0T} = \Omega \cdot \sqrt{1 + \frac{Z_1^2}{4Z_2^2}}$, $\Omega = \sqrt{Z_1 Z_2}$
 $U_e >$ donde $Z_{0T} = 0,248$, $Z_2 = \frac{f}{f_c}$
 $Z_{0T} = \sqrt{\frac{1}{4}} \cdot \sqrt{1 + \frac{f^2}{4}} = \frac{WE}{4} \sqrt{1 + \frac{f^2}{4}}$
 $Z_e = EAC * W = 0$ or $Z_{0T} = S_{01} = 50 \Omega$
 Mao 19 3 y nolen 50n

$\Omega \sqrt{1 + \frac{Z_1^2}{4Z_2^2}} = \left(\frac{0}{\Omega}\right)^2 = \sqrt{1 + \frac{Z_1^2}{4Z_2^2}}$
 $1 + \frac{Z_1^2}{4Z_2^2} = 2.05$ $4 + \frac{Z_1^2}{Z_2^2} = 131 \text{€ y.}$
 EA

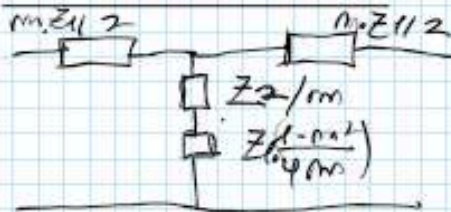
b-5w $\Rightarrow O = T = \frac{w^2 L C}{4} \Rightarrow \frac{w^2 L C}{Y} = 15$ (1)
 fo re Lodo, NN $\frac{Z_L \cdot \frac{1}{Z_C}}{4} = -VE 4:$
 $\left\{ \begin{array}{l} W_{ez} = \frac{1}{V_e} \text{ and } \Rightarrow \sqrt{LC} = \frac{2}{WC} \Rightarrow L = (7) E, \\ Z_{0T} = \sqrt{\frac{1}{4}} = S_{01} \Rightarrow L = S_{01}^2 \cdot C \end{array} \right.$
 $L = 4591 \text{ nH}$ > 15 6366 mF
 $S_{01}^2 \cdot C = \left(\frac{2}{WC}\right)^2 \cdot \frac{1}{C}$

Ejercicio6_Hoja_2

viernes, 24 of October

18:20

Celda m. derivada -



Pozo Bojo

$$W_{\infty} = \frac{W_C}{\sqrt{1-m^2}} \Rightarrow \sqrt{1-m^2} = \frac{W_C}{W_{\infty}}$$

$$\sqrt{1 - \left(\frac{W_C}{W_{\infty}} \right)^2} = m \Rightarrow m = \sqrt{1 - \left(\frac{20 \cdot 10^{-4} \text{ Hz}}{20 \cdot 10^{-4} \text{ Hz}} \right)^2}$$

$$m = 0,5527$$

$$Z_2' = \frac{Z_2}{m} + Z_1 \frac{(1-m^2)}{4m} = \left\{ \begin{array}{l} \frac{Z_2}{m} = \frac{1}{0,5527} \Rightarrow C' = C \cdot m = 3,5 \text{ nF} \\ C_2' = 0,366 \text{ nF} \cdot 0,5527 \end{array} \right.$$

$$Z_1' = m \frac{Z_1}{2} = m \frac{1}{2} \Rightarrow$$

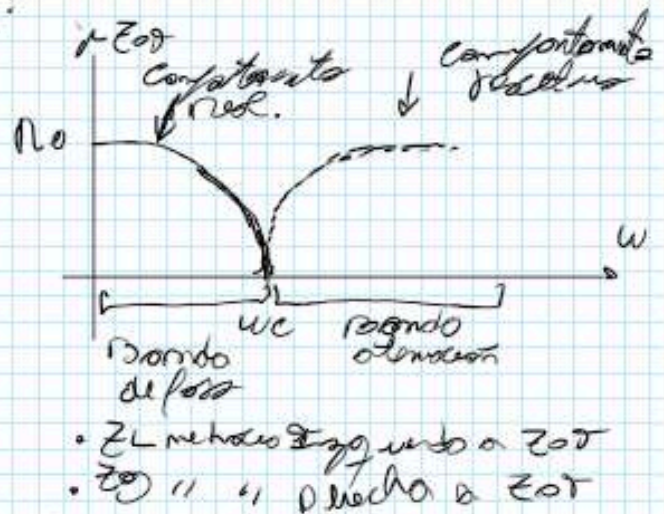
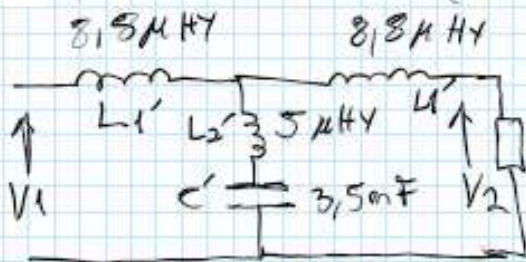
$$L_1' = m \cdot L = 0,5527 \cdot \frac{15,91 \mu\text{H}}{2}$$

$$L_1' = 8,74 \mu\text{H}$$

$$L_2' = 5 \mu\text{H}$$

it

celda m. derivada queda:



Obtenemos la matriz T de la celda considerando solo la

$$[T] = \begin{bmatrix} 1 + Z \cdot Y & Z \\ Y & 1 \end{bmatrix} \rightarrow \text{matriz T celda "Enk"}$$

$$[T] = \begin{bmatrix} 1 + 0,4 \cdot \frac{1}{3,5 \times 10^{-9} + \frac{1}{5 \times 10^{-6}}} & 0,4 \\ \frac{1}{3,5 \times 10^{-9} + \frac{1}{5 \times 10^{-6}}} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 + 0,4 \cdot \frac{1}{15,91 \times 10^{-6}} & 0,4 \\ \frac{1}{15,91 \times 10^{-6}} & 1 \end{bmatrix}$$

Exercises Hoja_3

wermes, October 2nd of 2025 18:58

$$[T] = \begin{bmatrix} \left(1 + \frac{\$L_1}{\$L_2 + \frac{1}{\$C'}}\right) \left(1 + \frac{\$L_1}{\$R_L}\right) + \frac{\$L_1}{\$R_L} & \left(1 + \frac{\$L_1}{\$L_2 + \frac{1}{\$C'}}\right) (\$L_1 + \$L_2 - 1) \\ \frac{1}{\$L_2 + \frac{1}{\$C'}} \left\{ \left(1 + \frac{\$L_1}{\$R_L}\right) + \frac{1}{\$R_L} \right\} & \frac{1}{\$L_2 + \frac{1}{\$C'}} \cdot \$L_1 + 1 \cdot 1 \end{bmatrix}$$

$$A = \left(1 + \frac{\$L_1}{\$L_2 + \frac{1}{\$C'}}\right) \cdot \left(1 + \frac{\$L_1}{\$R_L}\right) + \frac{\$L_1}{\$R_L}$$

$$A = \left(1 + \frac{\$L_1 \cdot \$C'}{\$L_2 \cdot \$C' + 1}\right) \left(1 + \frac{\$L_1}{\$R_L}\right) + \frac{\$L_1}{\$R_L}$$

$$A = 1 + \frac{\$L_1}{\$R_L} + \frac{\$L_1^2 \cdot \$C'}{\$L_2 \cdot \$C' + 1} + \frac{\$L_1^2 \cdot \$C'}{\$L_2 \cdot \$C' + 1} \cdot \frac{1}{\$R_L} + \frac{\$L_1}{\$R_L}$$

$$A = \frac{(\$L_2 \cdot \$C' + 1) \$R_L + (\$L_2 \cdot \$C' + 1) \cdot \$L_1 + \$L_1^2 \cdot \$C' \cdot \$R_L + \$L_1^2 \cdot \$C' \cdot \$L_1 + (\$L_2 \cdot \$C' + 1) \$L_1}{(\$L_2 \cdot \$C' + 1) \cdot \$R_L}$$

$$A = \frac{\$L_2 \cdot \$C' \cdot \$R_L + \$R_L + \$L_2 \cdot \$C' \cdot \$L_1 + \$L_1 + \$L_1^2 \cdot \$C' \cdot \$R_L + \$L_1^2 \cdot \$C' \cdot \$L_1 + \$L_1}{(\$L_2 \cdot \$C' + 1) \cdot \$R_L}$$

$$Az \quad \$L_1^3 (\$L_1 \cdot \$C' + \$L_1^2 \cdot \$C' + \$L_2 \cdot \$C' \cdot \$L_1) + \$L_1^2 (\$L_2 \cdot \$C' \cdot \$R_L + \$L_1 \cdot \$C' \cdot \$R_L) + \$L_2 \cdot \$C' + \$R_L$$

$$L_1 = 818 \mu H \quad (\$L_2 \cdot \$C' + 1) / \$R_L$$

$$L_2 = 3 \mu H, C' = 313 nF$$

$$A = \$L_1^3 5179 \cdot 10^{-19} + \$L_1^2 2415 \cdot 10^{-12} + \$L_1 176 \cdot 10^{-5} + 50$$

$$\$L_1^2 8175 \cdot 10^{-13} + 50$$

$$A(\omega) = \frac{\omega^3 5179 \cdot 10^{-19} - \omega^2 2415 \cdot 10^{-12} + \omega 176 \cdot 10^{-5} + 50}{-\omega^2 8175 \cdot 10^{-13} + 50}$$

Exerciseb_Hoja_4

Friday, 24 of October

2025

19:21

$$A(j\omega) = \frac{-j\omega^3 5,79 \cdot 10^{-19} - \omega^2 2,415 \cdot 10^{-12} + j\omega 1,176 \cdot 10^{-5} + 50}{- \omega^2 8,75 \cdot 10^{-13} + 50}$$

$$|A(j\omega)| = \frac{\sqrt{(-\omega^2 2,415 \cdot 10^{-12} + 50)^2 + (-\omega^3 5,79 \cdot 10^{-19} + \omega 1,176 \cdot 10^{-5})^2}}{\sqrt{(+\omega^2 8,75 \cdot 10^{-13} + 50)^2}}$$

$$\omega = 2\pi \cdot 1,2 \text{ MHz} \text{ para encontrar}$$

$$|A(j\omega = 2\pi \cdot 1,2 \text{ MHz})| = \frac{\sqrt{7619,57 + 13,334,8}}{0,257} = \frac{144,75}{0,257}$$

$$[T_o]_{(\omega = 2\pi \cdot 1,2 \text{ MHz})} = 563,23$$

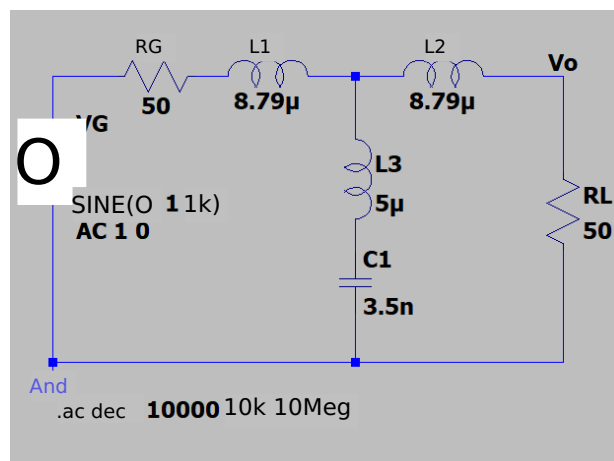
$$\frac{1}{|A(j\omega = 2\pi \cdot 1,2 \text{ MHz})|} = \frac{1}{563,23} = \frac{V_2}{V_1} = 1,775 \cdot 10^{-3}$$

$$[V_2]_{\text{dB}} = 20 \log(1,775 \cdot 10^{-3}) = -55 \text{ dB to } \omega = 2\pi \cdot 1,2 \text{ MHz}$$

goes

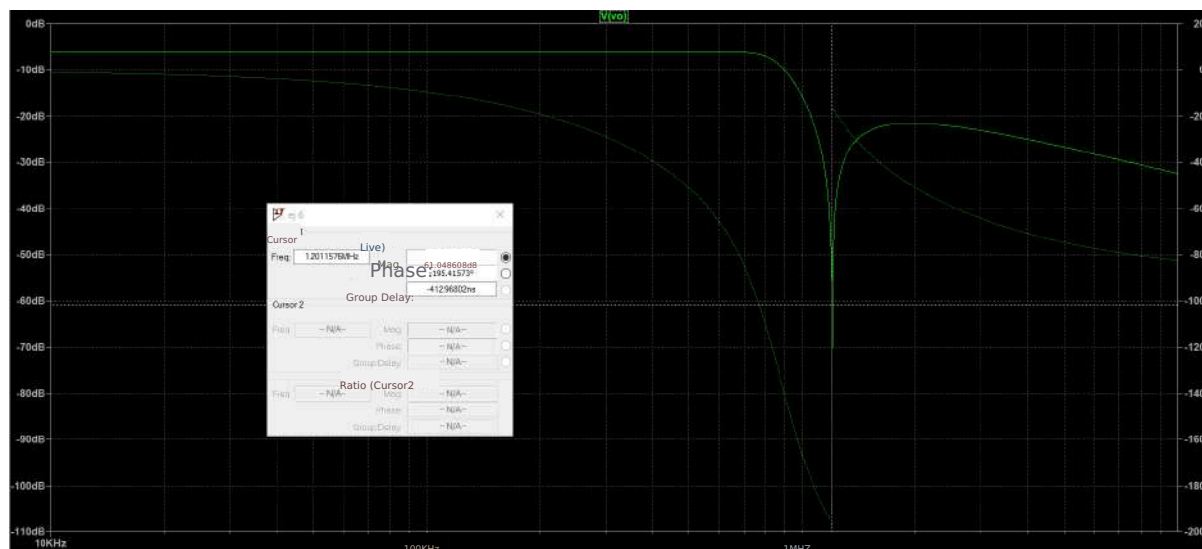
Conclusion: Con un único polo en -ordenada conseguimos cumplir con lo que se pide > 40dB, obteniendo todo mínimo en $f = 1,2 \text{ MHz}$

Simulation:



Circuit of exercise 6.

Plot 9

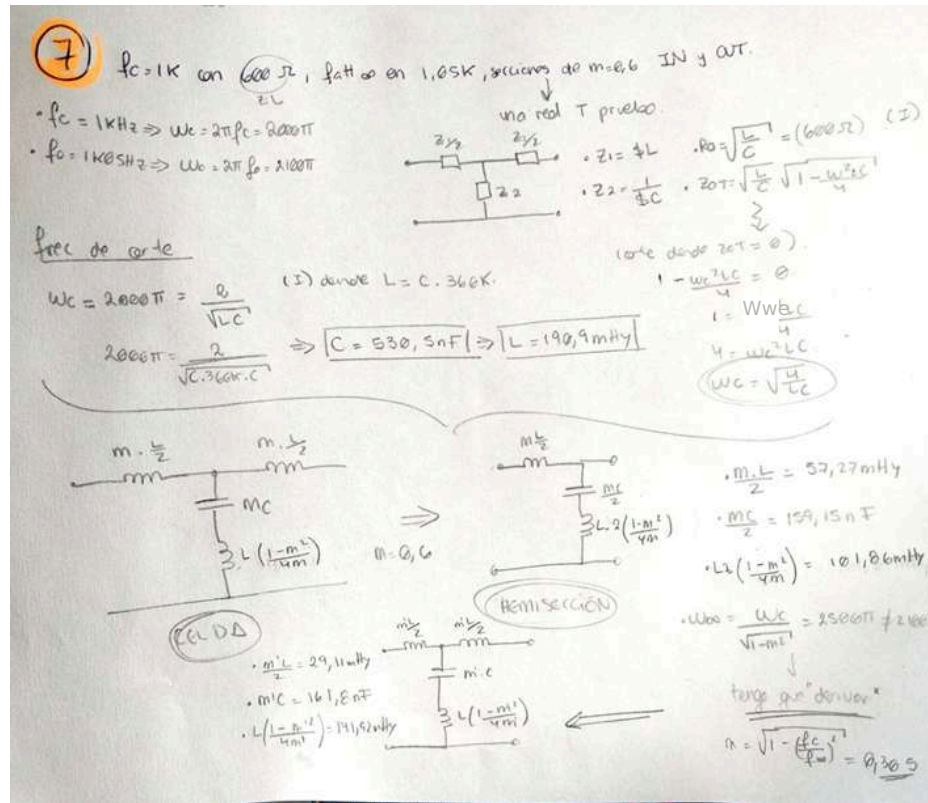


Graph 10

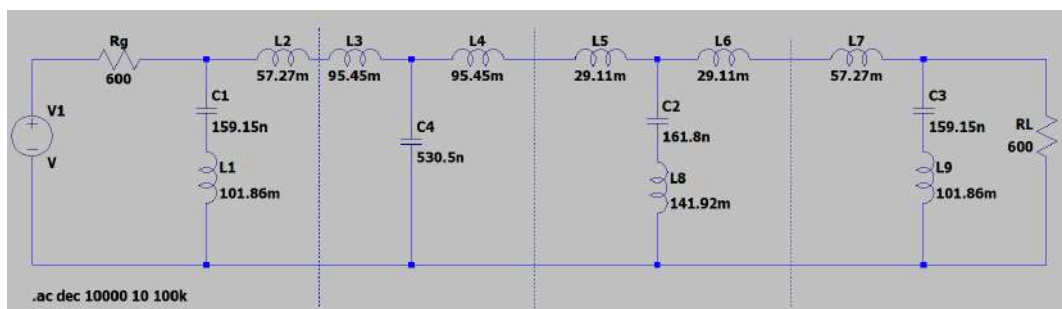
Frequency response of the voltage transfer of the circuit from exercise 6.

Exercise #7

Obtain a filter with a cutoff frequency of 1 kHz loaded with 600 ohms. To have good discrimination, it is desired to obtain an infinite attenuation frequency for some cell at 1.05 kHz. To properly match, sections with $m=0.6$ at the input and output are requested.

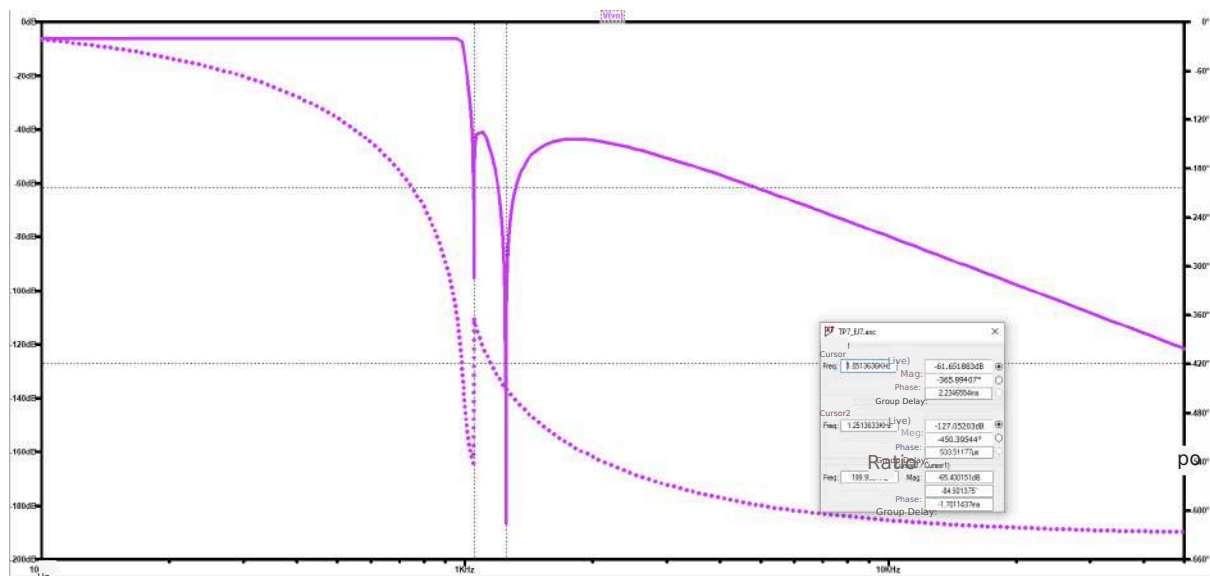


Simulation:



Circuit of exercise 7.

Graph



Plot 12

Frequency response of the voltage transfer function of the circuit from exercise 7.

Exercise #8

Exercises Sheet 1

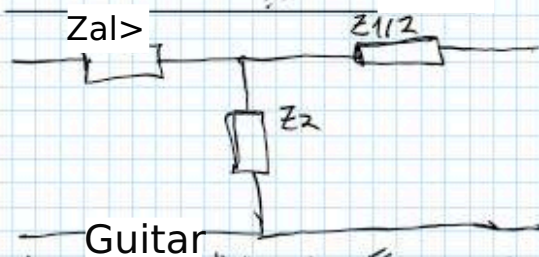
Sunday, October 12, 2025

11:56

Exercise 8. Given a K-cte type T structure, obtain the components to build a filter with the following specifications:

- Immersed in a system of 6000 ohms
- Cutoff; 10GHz
- Cero de transmission: 9.5GHz $\rightarrow$ se coloca en banda de atenuación

Estructura K-cte LY



Um. "Amplificador" en una de
2 pines, amplifica en un solo
- f

jar pots = -7 = No. fono

De los datos, se
filtro para el
 $\Rightarrow Z1/2$ the Pair
do 22

to Were Aa, el 0 de transmission
 $\Rightarrow$ se usa
 $\Rightarrow Z > 2$ Ab 9 222

10) Comenzamos con el K-cte

ez 1 $\frac{1}{2\sqrt{LC}}$ ① $\Rightarrow \sqrt{LC} = \frac{1}{\omega c_2}$ "az [a Y $\frac{2}{c}$]

0.4 T = More .C

$$\Rightarrow \sqrt{\frac{1}{LC}} = \omega c_2 \Rightarrow C = \frac{1}{\omega^2 L}$$

Camc = 83,33 Feny

L = 30 mH

Definición:

= z > ALLI Foc

ño nr! ml e NE feat >
Boo fo (e)

domingo, 12 from October 2025 12:04

$166,66 \cdot 10^{-15} \text{ F}$ $166,66 \cdot 10^{-15} \text{ F}$ $\frac{Z_1}{2} = \frac{1}{2C} \Rightarrow \frac{1}{5 \cdot 2 \cdot C} = \frac{1}{5 \cdot 166,66 \text{ fF}}$

 And
 $Z_2 = j\omega L = j \cdot 30 \text{ mH}$

$G_{eq} = \frac{1}{\sqrt{\frac{L' C'}{g}}}$
 $\frac{Z_1}{2} = \frac{1}{\sqrt{\frac{L' C'}{g}}}$
 $C'_{eq} = \frac{2C}{\omega}$

$ZC' = \frac{Z_1}{2} \left(\frac{1 - m^2}{4m} \right)$
 $ZL' = \frac{Z_2}{m}$
 = Series Components

" Same ^m components short
look to same old
transmission in Wos

$$Z' = \frac{1}{s} \cdot \left(\frac{1-m^2}{4m} \right) \Rightarrow \frac{1}{s} \cdot \frac{c+4m}{1-m^2}$$

$$\frac{.4 \text{ m} \cdot \text{C}}{1 \text{ m}^2}$$

$$Z_{\text{short}}' = \frac{Z_2}{m} = \frac{gL}{m} = g \cdot L', \quad L' = \frac{L}{m}$$

$$\omega_{os} = \frac{l}{\sqrt{\frac{l}{m} \cdot \frac{4\pi m \cdot c}{1-\beta^2}}} = \frac{l}{\sqrt{\frac{4 \cdot l \cdot c}{1-\beta^2}}}, \text{ then } \omega c^2 = \frac{l}{(2\sqrt{LC})^2} = \frac{l}{4LC} = \frac{l}{\omega c^2} \text{ "Le"}$$

$$w_{\text{res}} = \frac{l}{\sqrt{\frac{d/wc^2}{1-\beta^2}}} = \frac{1}{\sqrt{\frac{1}{wc^2} \frac{1}{1-\beta^2}}} = \frac{A}{\sqrt{\frac{1}{wc^2} \frac{1}{1-\beta^2}}}$$

$W_{\text{iso}} = W_c \cdot \sqrt{1 - \sin^2 \alpha}$ → Prozessalter
Influente → Resümee

Definieren "rm"
 $\frac{W_{ss}}{W_c} = \sqrt{1 - r_m^2} \Rightarrow \sqrt{1 - \left(\frac{W_{ss}}{W_c}\right)^2} = r_m$ → sozusagen
wichtige Aussagen

Ejercicio8_Hoja_3

domingo, 12 de octubre de 2025 12:31

Despejamos γ

$$\frac{W_{00}}{W_e} = \sqrt{1 - \gamma^2} \Rightarrow \sqrt{1 - \left(\frac{W_{00}}{W_e}\right)^2} = \gamma$$

o por altos
importante en la dñon

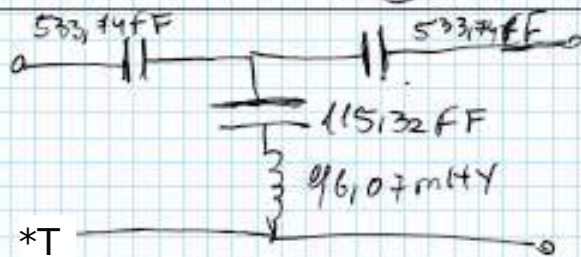
$$\gamma = \sqrt{1 - \left(\frac{9,5644}{106,47}\right)^2} = 0,31225$$

$$C'_{\text{shunt}} = \frac{2C}{1 - \gamma^2} = \frac{2 \cdot 83,33 \text{ fF}}{1 - 0,31225^2} = 533,74 \cdot 10^{-15}$$

$$C'_{\text{shunt}} = \frac{4m \cdot C}{1 - \gamma^2} = \frac{4 \cdot 0,31225 \cdot 83,33 \text{ fF}}{1 - 0,31225^2} = 115,32 \text{ fF}$$

$$L'_{\text{shunt}} = \frac{L}{\gamma} = \frac{30 \text{ mH}}{0,31225} = 96,076 \text{ mH}$$

lo alto m de modo como como

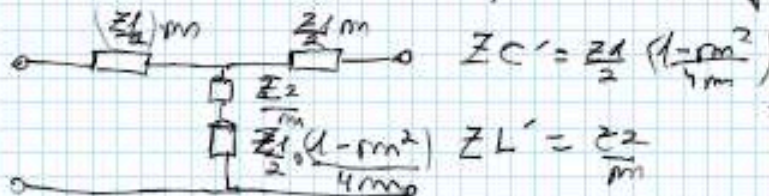


*T

2.2) Continuamos con la normalización

Existen de diseño $m = 0,6$, continuamos de m y por altos

$$\Rightarrow W_{00} = W_e \cdot \sqrt{1 - m^2}, Z_{0T} = N_0 \cdot \sqrt{1 + \frac{Z_1}{4Z_2}}$$



$\Rightarrow$ Tomamos Z_1 y Z_2 para hacer la normalización

$$C'_{\text{shunt}} = \frac{4m \cdot C}{1 - m^2} = \frac{4 \cdot 0,6 \cdot 83,33 \text{ fF}}{1 - 0,6^2} = 312,4875 \cdot 10^{-15} \text{ F}$$

$$L'_{\text{shunt}} = \frac{L}{m} = \frac{30 \text{ mH}}{0,6} = 50 \text{ mH}$$

$$C'_{\text{series}} = \frac{2C}{m} = \frac{2 \cdot 83,33 \text{ fF}}{0,6} = 277,77 \cdot 10^{-15} \text{ F}$$

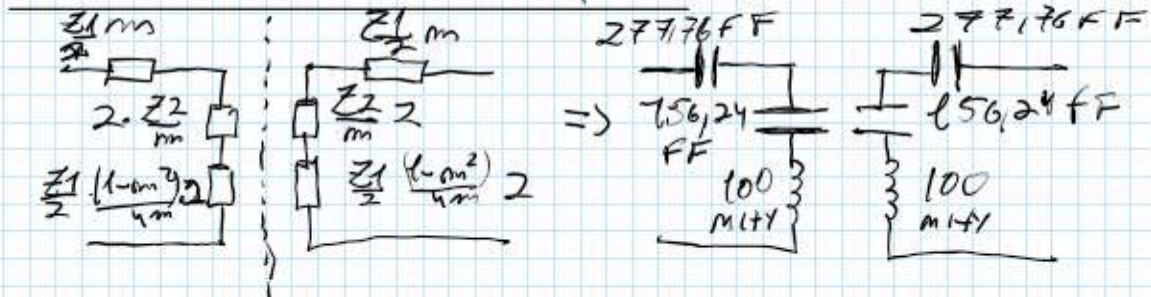
Exercises Sheet_4

domingo, 12 of October

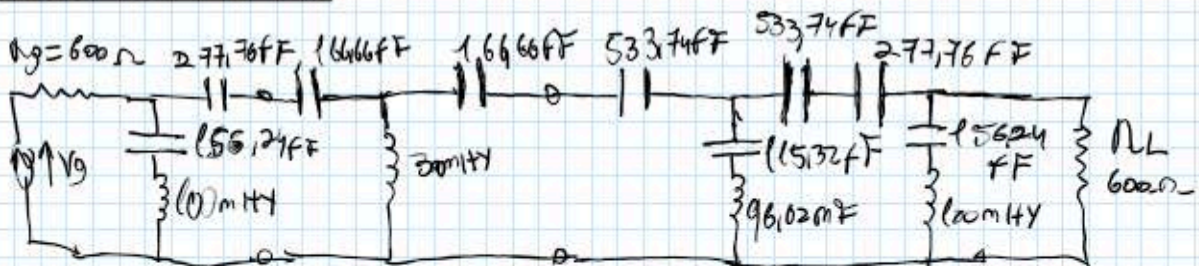
2025

12:46

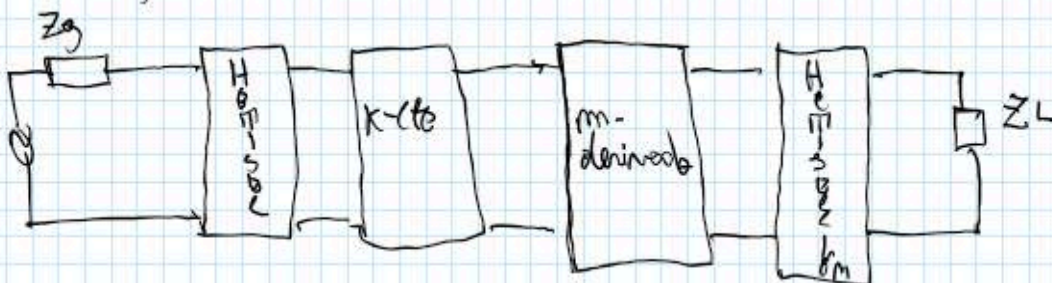
el circuito de hemisección queda:



Circuito total



Enbloques



- El "m" derivado genera ceros de transmisión $\omega_{00} = 9.564 \text{ rad/s}$
- la hemisección, también introduce un cero de transmisión pero un ω_{00} tal que $m = 0$, b. permite mantener el Z_{01} de todo el bloque de fase
- los $K < 1$ reducen la banda de paso y atenúan pero no introduce "0" de transmisión

Simulation:

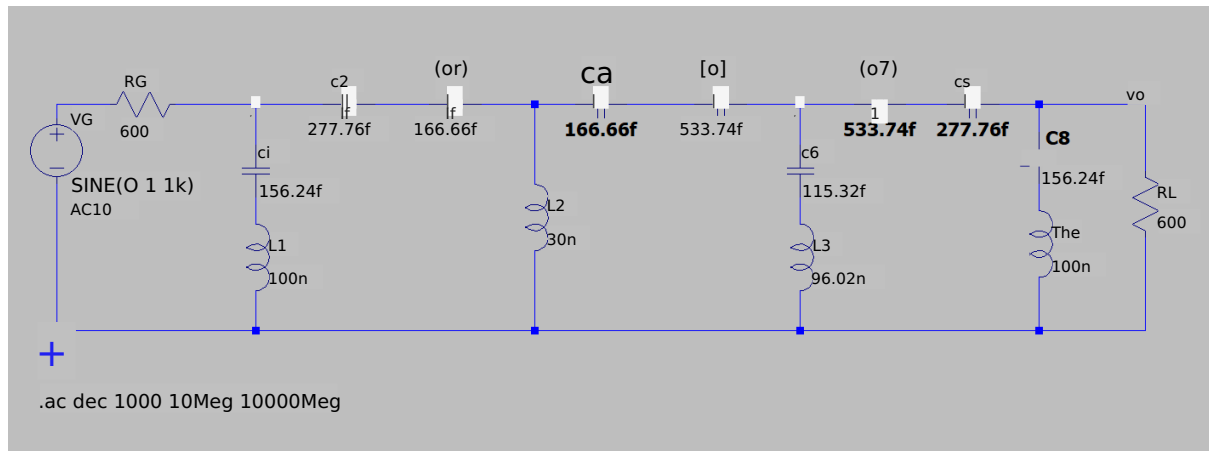


Figure 13

Circuit of exercise 8.

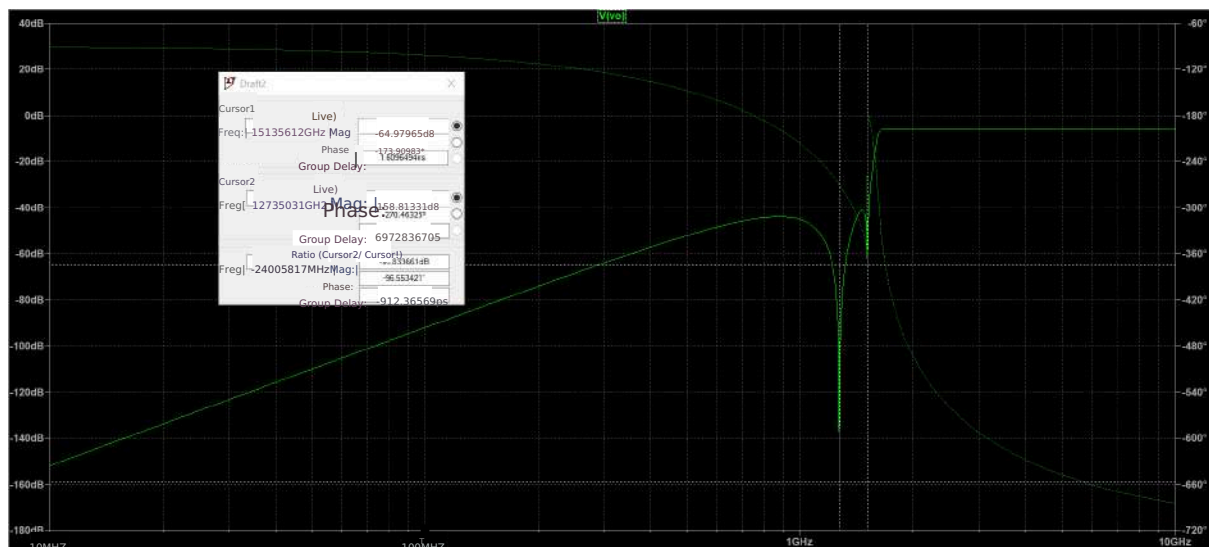


Figure 14

Frequency response of the voltage transfer of the circuit from exercise 8.

Exercise #9

Exercise 9, Design a low-pass filter with K-constant with a cutoff frequency of 2kHz loaded with its characteristic impedance of 600 ohms. Simulate the proposed design and verify its operation.

177 do, the KLe

2/7

Use

Notes: $RCE = 2KHz$, $R_0 = 600\Omega$

KN cut-off frequency

Hemispherical

en

$C = 265,25mF$, $L = 92,5mH$

La

TT ZA

pe u 65mH

50 - 43m 7

mi

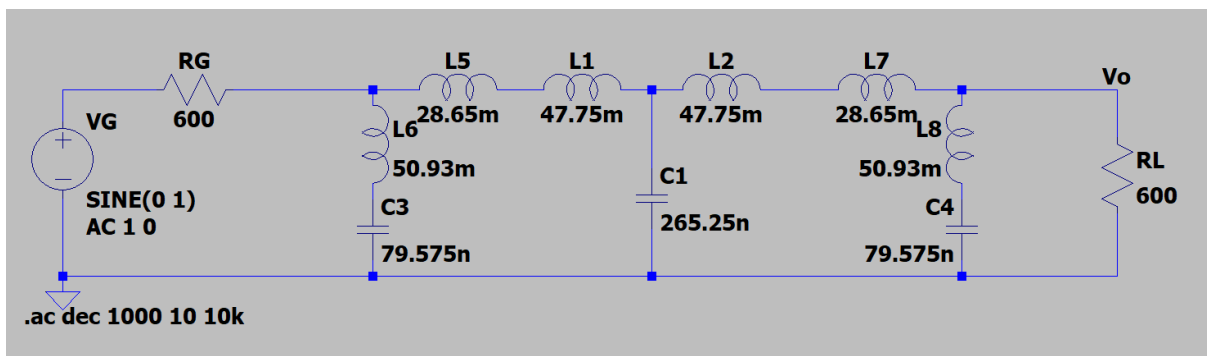
79,5

57mF

$C \Rightarrow \sqrt{\frac{4}{N} \frac{1}{u_{We}^4} \cdot \frac{1}{m^4}} = \sqrt{\frac{4}{2 \cdot 2KHz} \cdot \frac{1}{600^2} \cdot \frac{1}{m^4}} = 265,290$

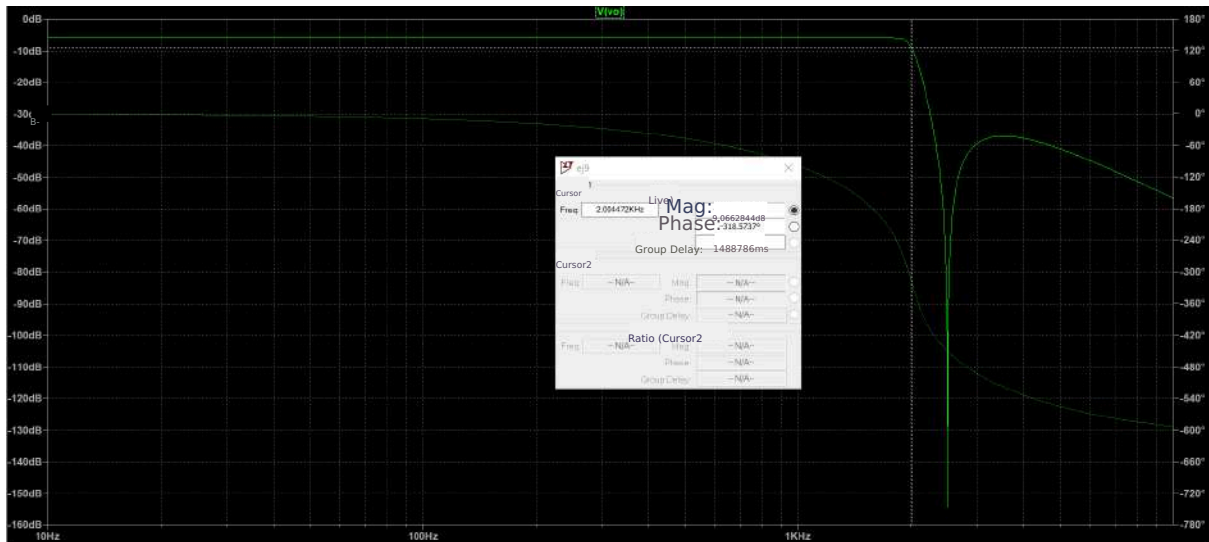
$L = \frac{4}{(2\pi \cdot 2KHz)^2} \cdot \frac{1}{265,25mF} = 95,5mH$

Simulation:



Graph 15

Circuit of exercise 9.



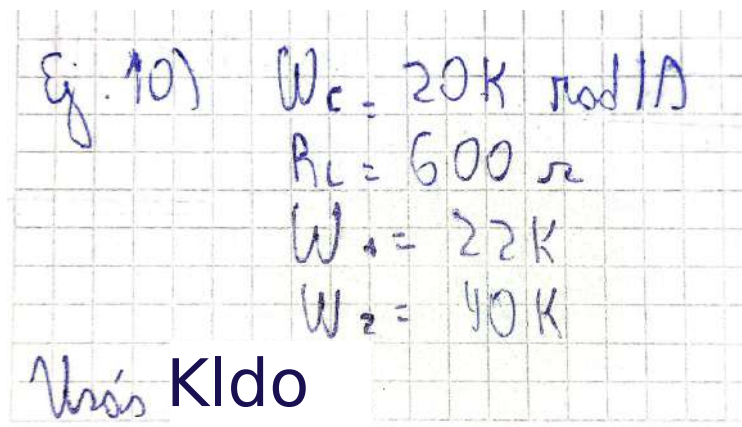
Frequency response of the voltage transfer of the circuit in exercise 9.

Graph 16

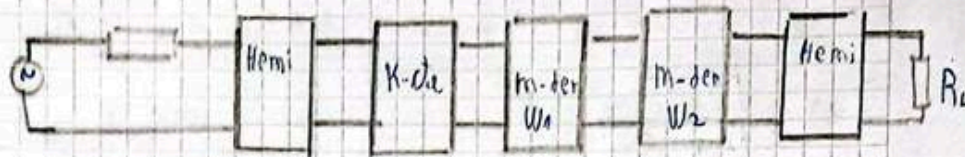
Exercise #10

Exercise 10. Design a Low-Pass filter that satisfies the following specifications:

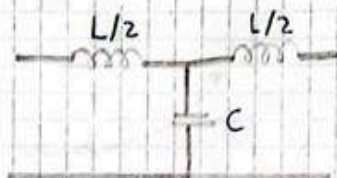
- cutoff = 20K rad/s
- $R_L = 6000$ ohms
- Infinite attenuation in:
 - or $\omega_1 = 22\text{Krad/s}$
 - or $\omega_2 = 40\text{Krad/s}$
 - or $\omega_0 = \infty$
- The filter must be impedance-matched at input and output in the passband.
- Simulate the proposed design and verify its operation.



Eg. 10) $\omega_c = 20\text{K rad/s}$
 $R_L = 600 \Omega$
 $\omega_1 = 22\text{K}$
 $\omega_2 = 40\text{K}$
This Kldo



Planteas celda K-de:



$$R_0 = \sqrt{\frac{L}{C}}$$

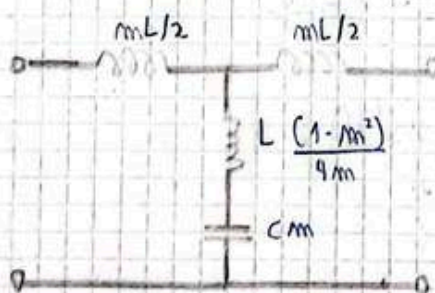
$$W_c = \frac{2}{\sqrt{LC}} \longrightarrow C = 166,6^{to} \text{ nF}$$

$$L = 60 \text{ mH}$$



Calcul's m para $W_\infty = W_1$ $W_1 = \frac{W_c}{\sqrt{1-m^2}} \longrightarrow m = 0,417$

$$W_1 = 22 \text{ Krad/s}$$

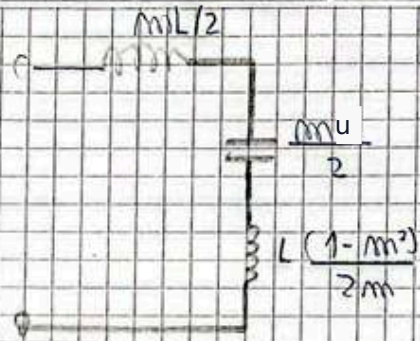


En $W_\infty = W_2$:

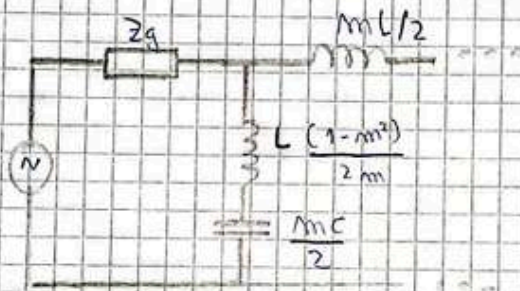
$$W_2 = 40 \text{ Krad/seg}$$

$$W_2 = \frac{W_c}{\sqrt{1-m^2}} \longrightarrow m = 0,866$$

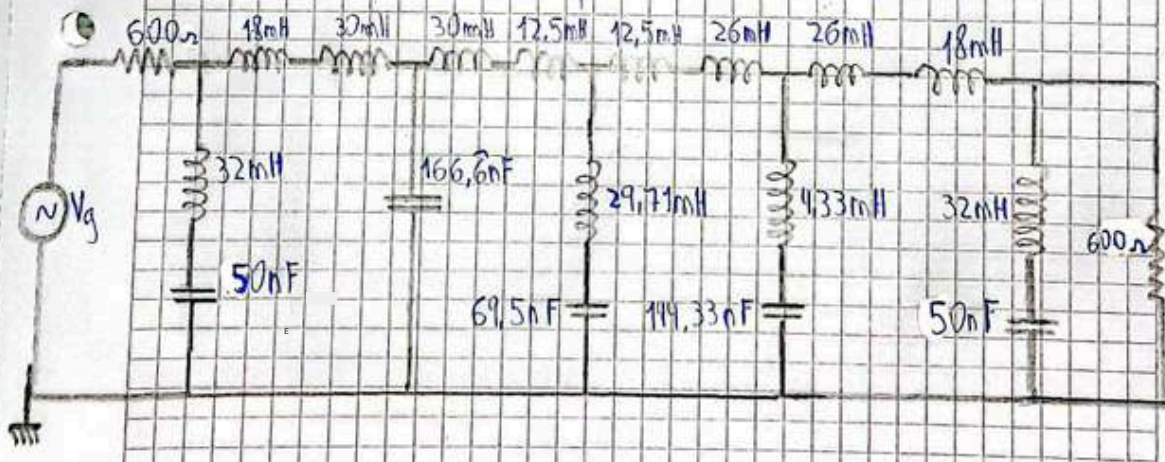
En la hemi-recta:



$$W_{\infty} \Big|_{m_e = 0.6} = 25 \text{ K } \frac{\text{rad}}{\text{s}}$$



Finalmente, el circuito completo se vería así:



Simulation:

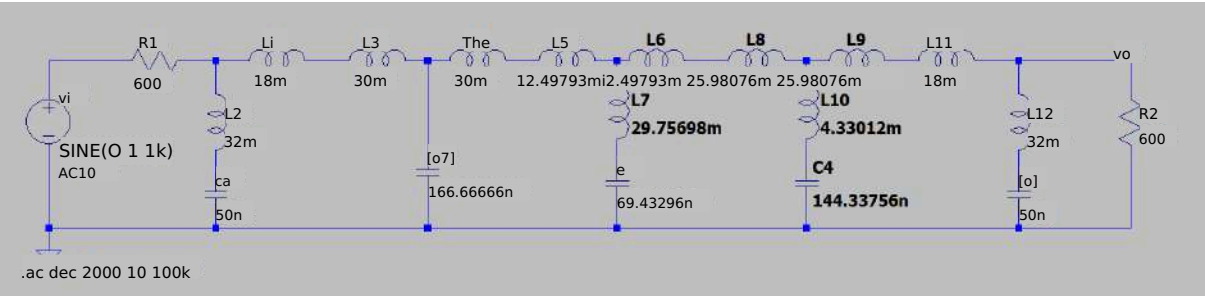


Figure 17

Circuit of exercise 10.

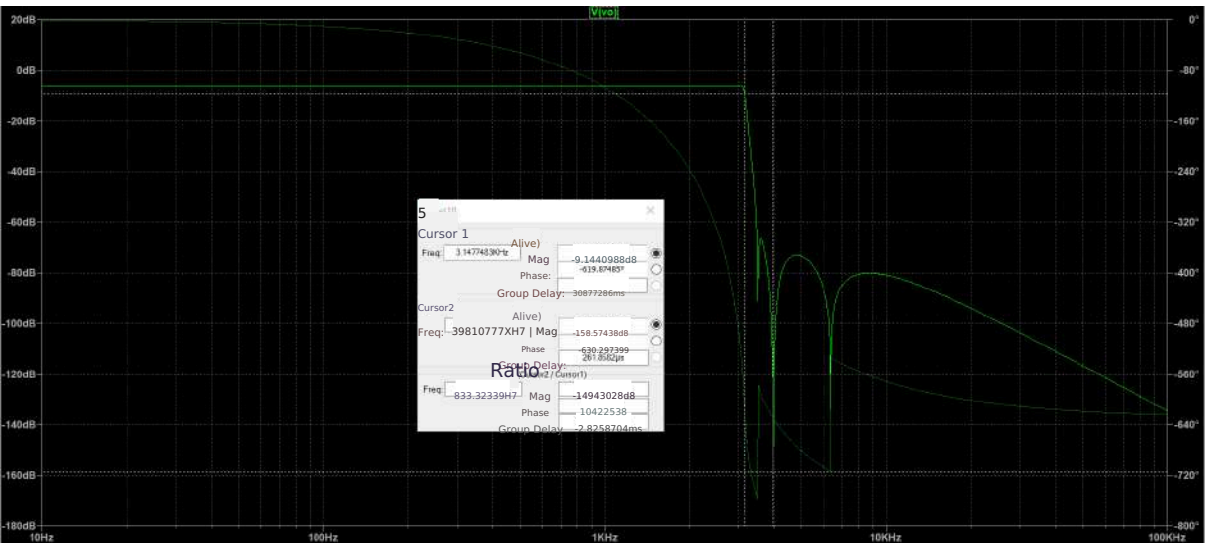


Figure 18

Frequency response of the voltage transfer of the circuit from exercise 10 number 1.

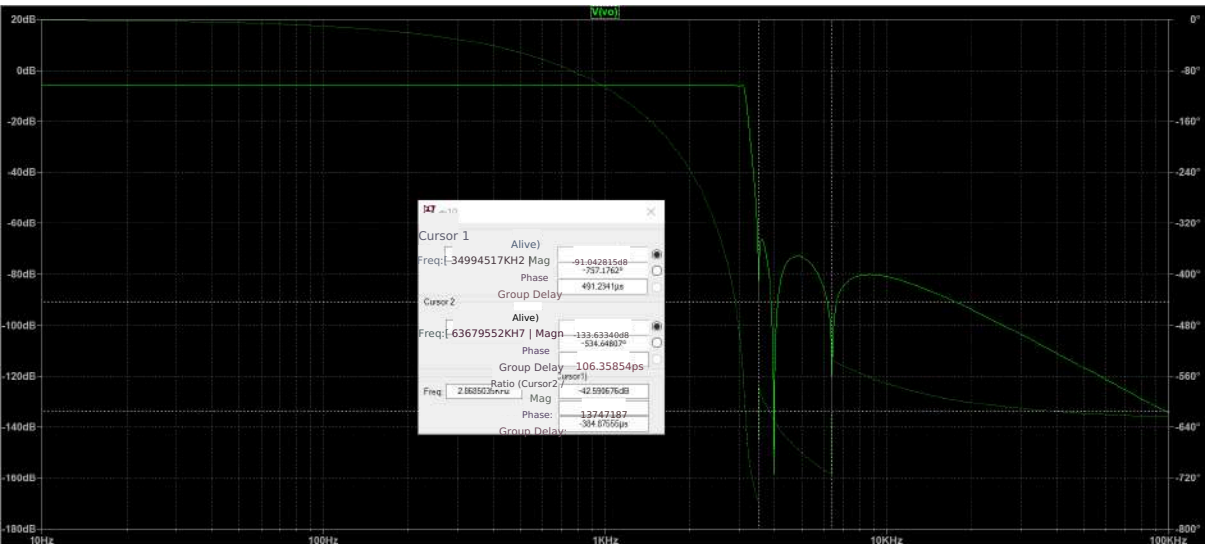


Gráfico 19

Frequency response of the voltage transfer of the circuit from exercise 10 number 2.

Exercise #11

Exercise 1. Sheet 1

Sunday, October 17, 2025

2311

Exercise 11. Design a broadband attenuator that allows 30 dB attenuation interleaved in a 75 ohm coaxial cable,

Death $[T] = \begin{bmatrix} NE \text{ aa } \cosh(\alpha) & \text{And } \sqrt{Z_{12}} \sinh(\alpha) \\ -\frac{f}{\sqrt{Z_{11} \cdot Z_{12}}} \sinh(\alpha) & \sqrt{\frac{Z_{12}}{Z}} \cosh(\alpha) \end{bmatrix}$

6 K Alo 14 α of $\cosh(\alpha)$ (27% Ando $\beta = 0$
requiere MM $\Rightarrow a$ $\beta = 0$
Resistencia $\Rightarrow \beta = 0$. $\beta = 0$
emenda

0 > dot $\rightarrow \alpha_{mp} = 3049.268 = 3.46 \text{ NP}, \alpha = AX$

Paras Id: $\cosh(\alpha) = 15.644, \sinh(\alpha) = 65.22, 00 = 156$

e Igualamos of lares 17005 α or fe $\beta = 0$ 17 e Eo
dada como $\sinh(\alpha) = 65.22$ Cone $\beta = 0$
adapta Loria $\sinh(\alpha) = 65.22$ $\beta = 0$
solido.

e $Z_{11} = Z_{12} = 75$ mA from $[T] = \begin{bmatrix} 15.86 & 11.8745 \\ 0.241 & 15.86 \end{bmatrix}$
a and AD V_i
a $V_1 = V_2 \cdot A + (-I_2) \cdot Z_{12}$ $\cosh(\alpha) = 15.86$
La = $V_2 \cdot C + (-I_2) \cdot Z_{22}$ $\sinh(\alpha) = 65.22$
Chaos) $V = \cosh(\alpha) \cdot V_2 + \sinh(\alpha) \cdot I_2 \cdot Z_{12}$
 $A = A$

$I_2 = 0 \Rightarrow MZA |$ V_1 ETRE
 V_2 $V_a = I_1 \cdot Z_{21}$

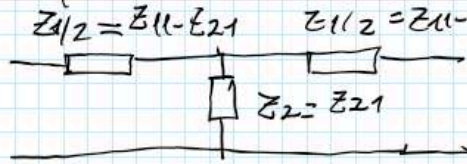
$I_2 = 0 \Rightarrow E_i$ $E_1 \Rightarrow$ Vez $Z_{12} \rightarrow 0$ $\left| \frac{I_1}{I_2} = \frac{1}{Z_{21}} = \right|$

" Can A € 90 $\sinh(\alpha)$ lighthouse He $\sinh(\alpha)$

Ejercicio11_Hoja_2

Sunday, 12 of October 2025 23:26

• Proponemos circuito atenuador tipo T

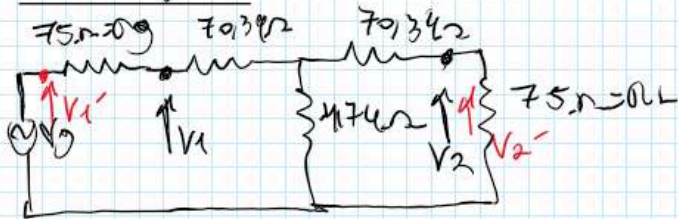


$$A = 15.86 \Rightarrow Z_{11} = A \cdot Z_{21} = 7513 \Omega$$

$$C_c = 0.211 = \frac{1}{Z_{21}} \Rightarrow Z_{21} = 4.74 \Omega$$

$$Z_{11} - Z_{21} = 7934 \Omega$$

circuito final



Exercise11 Hoja_3

Wednesday, 10 October 2025 13:00

Substitución

and En un cuadripolo simétrico tipo atenuador, podemos escribir su matriz:

$$\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} \cosh(\gamma) & Z_0 \sinh(\gamma) \\ \frac{1}{Z_0} \sinh(\gamma) & \cosh(\gamma) \end{pmatrix} \begin{pmatrix} V_2 \\ I_2 \end{pmatrix}, \quad \gamma = \alpha + j\beta$$

• Donde $A = \cosh(\gamma)$, $B = Z_0 \sinh(\gamma)$, $C = \frac{1}{Z_0} \sinh(\gamma)$

• Este $A = \cosh(\gamma)$ No es la ganancia V_1^2 / V_2^2 puesto que esto mide, solo cuando tenemos soldado a un lado "electrodo" $I_2 = 0$

• En atenuación, $A = \cosh(\gamma)$ es un parámetro más.

• Por otro lado, se demuestra que, al tener este cuadripolo con Z_L y corriente entrando al punto 2, entonces

$$\text{La ganancia de tensión es } \frac{V_1}{V_2} = \frac{1}{A + \frac{B}{Z_L}}$$

• Si el circuito queda a este cuadripolo y atenuador está adaptado $\Rightarrow$ se cumple que $Z_L = Z_0$, $\beta = 0 \Rightarrow \gamma = \alpha$ fa sea resistencia pura, lo que es solo característica de los atenuadores

$$\Rightarrow \text{quedo que: } \frac{V_2}{V_1} = \frac{A}{\cosh(\alpha) + \sinh(\alpha)} = e^{-\alpha} = \frac{1}{A}$$

• Vamos que La atenuación en tensión en circuitos adaptados es: $\bar{e}^{-\alpha} = 10^{-A(dB)/20}$

$$\bullet \text{ Queremos } A(dB) = 30 \text{ dB} \Rightarrow \bar{e}^{-\alpha} = 10^{-30/20} \Rightarrow 0.03162$$

$$\bar{e}^{-\alpha} = 0.03162 \Rightarrow \ln(\bar{e}^{-\alpha}) \Rightarrow -\alpha = \ln(0.03162) \Rightarrow \alpha = 3.46 \text{ NP}$$

$$\bullet 1 \text{ NP} \rightarrow 8.6859 \Rightarrow 3.46 \text{ NP} \cdot 8.6859 \approx 30 \text{ dB}$$

and Queremos $\frac{1}{\cosh(\alpha)} = \frac{1}{A} \approx 0.031 \Rightarrow -24 \text{ dB}$

Donde los 6dB restantes para llegar a los 30dB son donde se define a que $\cosh(\alpha) = \frac{e^{\alpha} + e^{-\alpha}}{2} \approx LE$

Ejercicio11_Hoja_4

Wednesday 20 of October of 2025 14:15

Veremos que el orden de $\cosh(x)$ es el mismo que el de e^x lo que nos da un factor Ale "2" que divide a $\frac{1}{2}$ por lo que
 $Jem \frac{1}{\cosh}$

• De concluye que

$$\text{ganancia adoflora} \Rightarrow \frac{V_o}{V_i} = e^{-\alpha} = 10^{-A_{dB}/20}$$

$$\Rightarrow \text{No adoflora} \Rightarrow \frac{V_o}{V_i} = e^{A} = e^{\cosh A}, \beta = 2 \alpha \sinh(\alpha)$$

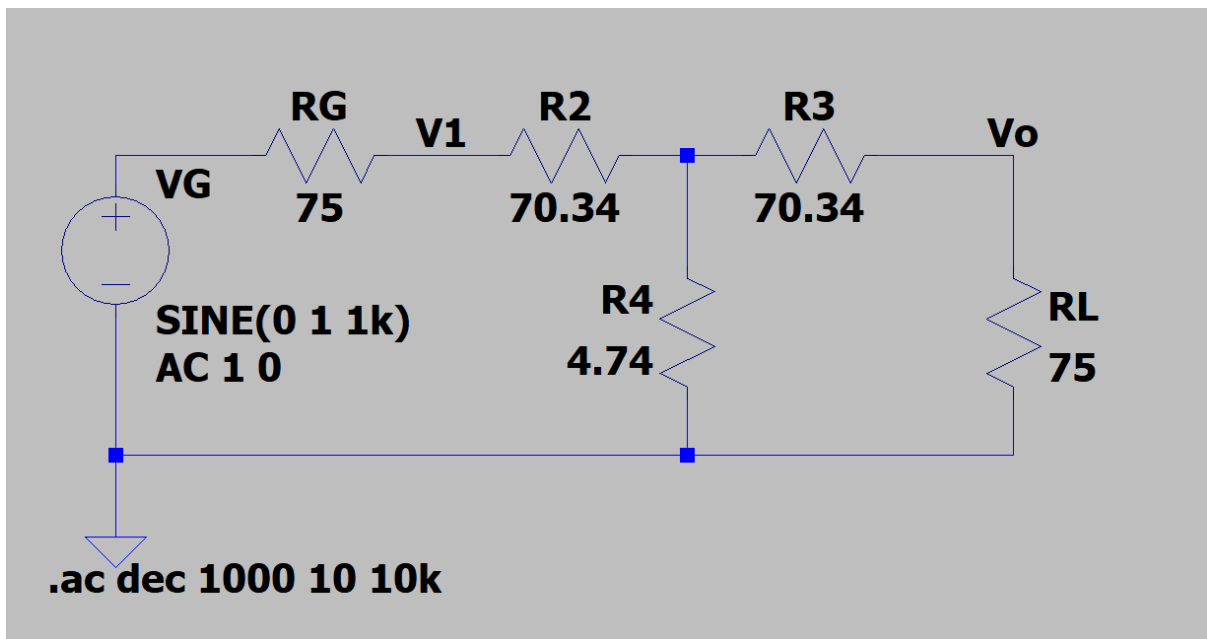
• With $R_2 = R_3 = 70.34 \Omega$, $R_4 = 4.74 \Omega$

$$\Rightarrow \left| \frac{V_o}{V_i} \right| \approx 0.0362 \rightarrow \text{adoflora} \Rightarrow 30 \text{ dB}$$

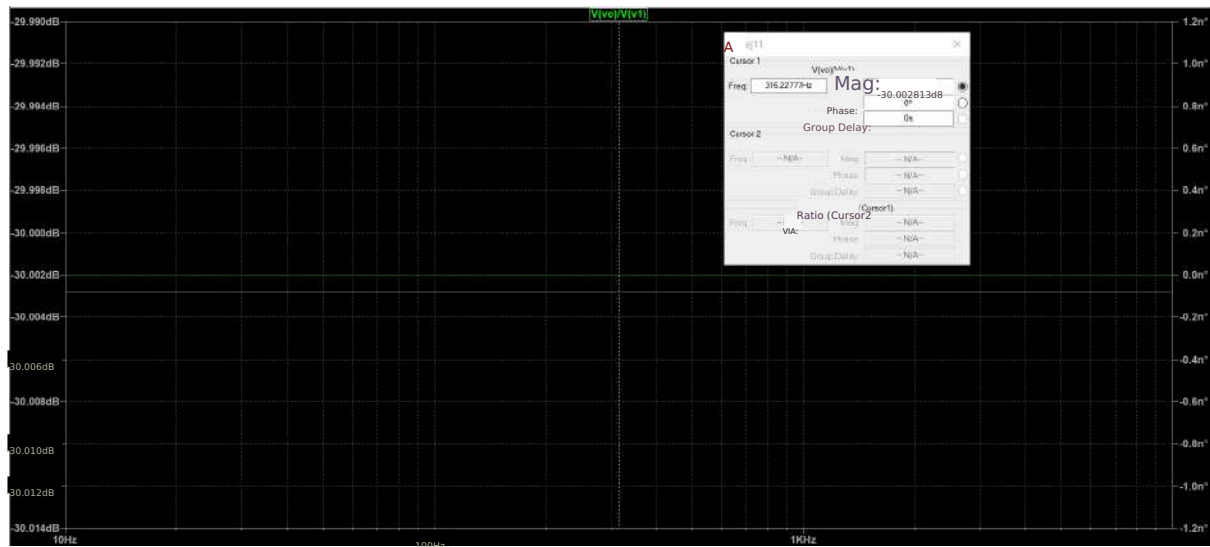
$$\Rightarrow \left| \frac{V_i}{V_{in}} \right| \approx 0.5 \rightarrow \text{de la fuente} \Rightarrow 6 \text{ dB} \rightarrow \text{siempre}$$

$$\Rightarrow \text{En total, and} \Rightarrow \text{Ganancia } 36 \text{ dB}$$

Simulation:



Circuit of exercise 11.



Graph 21

Frequency response of the voltage transfer of the circuit from exercise 11.

Exercise #12

Ejercicio 12 Page_1

Sunday, October 17, 2075

23:56

Exercise 12. Design a broadband matcher that allows interconnecting by adapting two cables coaxiales. In its input it should allow connection of a 75 ohm impedance coaxial cable characteristic and at its output one of 50 ohms impedance. It must attenuate 5.72 of.

Verify the matching, the attenuation of the resulting design and draw conclusions,

Planteo $[T] = \begin{bmatrix} \text{NE 7072} \\ \frac{1}{\sqrt{Z_{11} \cdot Z_{12}}} \sqrt{\frac{Z_{22}}{Z_{21}}} \end{bmatrix}$

$\alpha = \alpha + j\beta$ In conclusión, What (27 Ando $\Rightarrow \beta = E$

Además (4 there $\Rightarrow$ pa $\Rightarrow$ $\beta = 0$. Pa Aer Aga

Ad a hor, 97 254 cm^2 - see $= 265 \text{ MP}$, $\alpha = Xx$
677

ocurre ϕ de $Z_{11} = 75 \Omega$, $Z_{12} = \text{Are}$
• la matriz $[T] = \begin{bmatrix} SS & 43,34 \\ 0,0156 & 4 \end{bmatrix}$.

$V_1 = V_2 \cdot A + (-I_2) \cdot PH \text{ tr}$
 $Ta = Na \cdot \epsilon + (-I_2)$
Ecuación $\begin{cases} V_1 = I_1 \cdot Z_{11} + I_2 \cdot Z_{12} \\ V_2 = I_1 \cdot Z_{21} + I_2 \cdot Z_{22} \end{cases}$

$I_2 = 0 \Rightarrow \frac{V_1}{V_2} = A > \frac{V_A = I_1 \cdot Z_{11} + \text{or}}{N_C = I_1 \cdot Z_{21}} = \frac{V_A}{Y_i \cdot Z_{22} - A}$

$I_2 = 0 \Rightarrow \frac{I_1}{Y_2} = E, \Rightarrow V_A \quad I_1 \cdot Z_{21} + 0 \Rightarrow \boxed{\frac{I_1}{Y_2} = \frac{1}{Z_{21}} = C}$

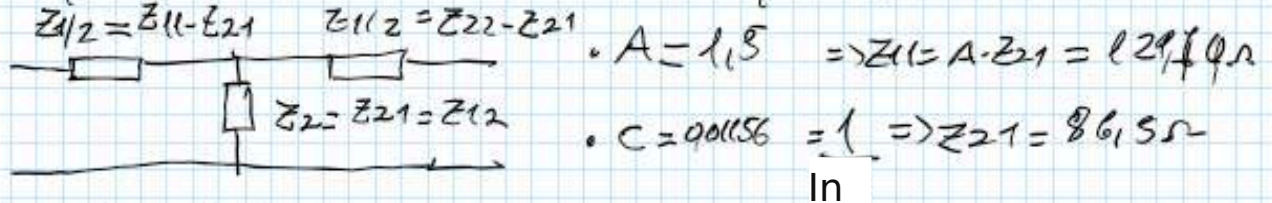
Dd: E ZH W-0

Con A, ϕ D suficiente para retirar los volantes del circuito

Ejercicio12Sheet 2

Monday 30 of October of 2025 00:03

Problema circuito atenuador tipo T



$$Z_{1/2} = Z_{11} - Z_{21}$$

$$Z_{11/2} = Z_{22} - Z_{21}$$

$$Z_2 = Z_{21} = Z_{12}$$

$$A = 1.5 \Rightarrow Z_{11} = A \cdot Z_{21} = 129.4 \Omega$$

$$C = 0.0156 = \frac{1}{Z_{21}} \Rightarrow Z_{21} = 86.5 \Omega$$

In

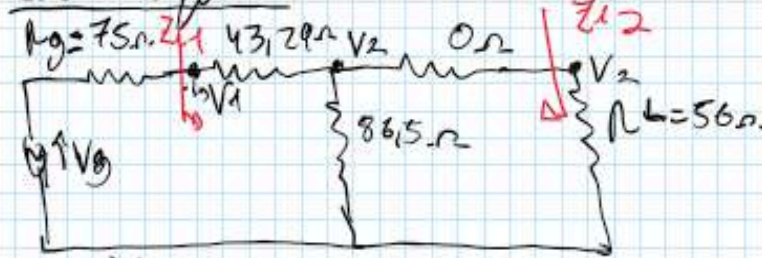
$$D = A = y \quad Z_{22} = D \cdot Z_{21} = 86.5 \Omega$$

$$Z_{11} - Z_{21} = 43.2 \Omega$$

$$Z_{22} - Z_{21} = 0 \Omega \text{ cable}$$

$$Z_{21} = Z_{12} = 86.5 \Omega$$

Circuito final.



Verificación

$$Z_{11} = 43.2 \Omega + (86.5 \parallel 56 \Omega)$$

$$Z_{11} = 74.95 \Omega$$

$$Z_{i2} = (75 \Omega + 43.2 \Omega) \parallel 86.5 \Omega$$

$$Z_{i2} = 49.97 \Omega$$

$$\text{Time } \frac{V_2}{V_1} = \frac{86.5 \parallel 56 \Omega}{(75 + 43.2) + (86.5 \parallel 56 \Omega)} = \frac{31.6849}{75 + 43.2 + 31.6849}$$

$$V_2 = V_1 \cdot 0.211268$$

$$20 \log \left(\frac{V_2}{V_1} \right) = 20 \log \left(\frac{V_1 \cdot 0.211268}{V_1} \right) = -13.1 \text{ Sello} \rightarrow \text{Atenuación obtenida}$$

Veremos que $\frac{L_{3/5}}{S_{1/2}} = 2.36 \Rightarrow$ Lo atenuación Lg with lo

Impedancias perdidas es 2.36 veces mayor o lo esperado. Este poly deberse a que $Z_{as} \quad Z_{i2}$

Simulation:

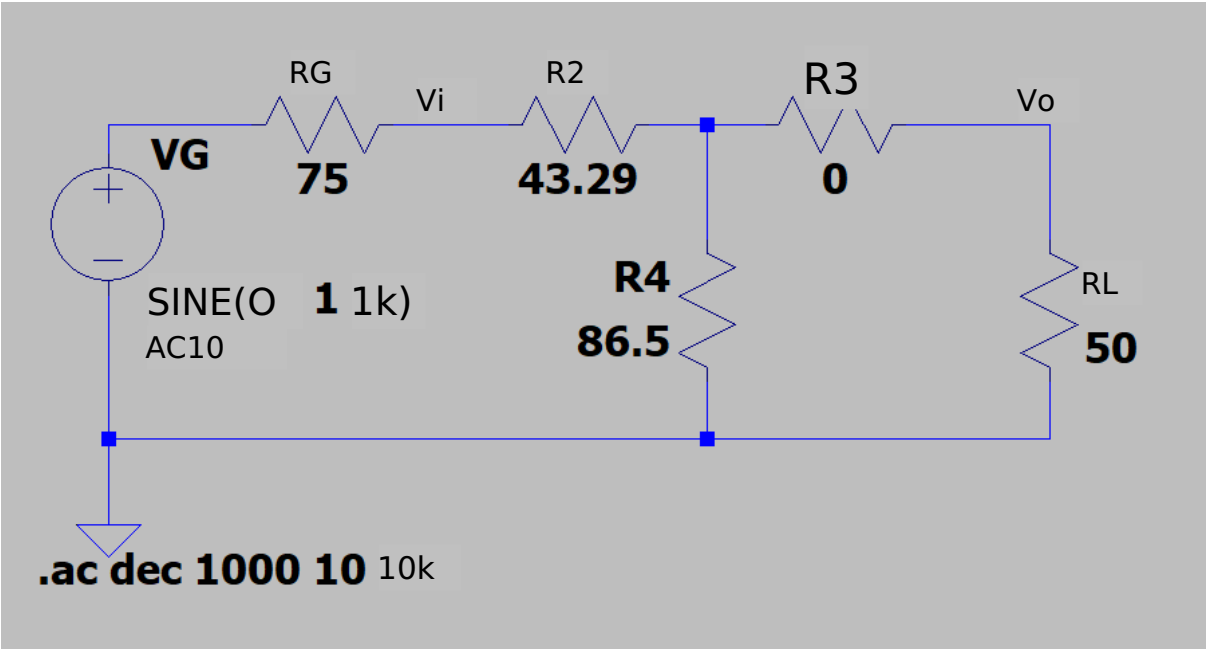
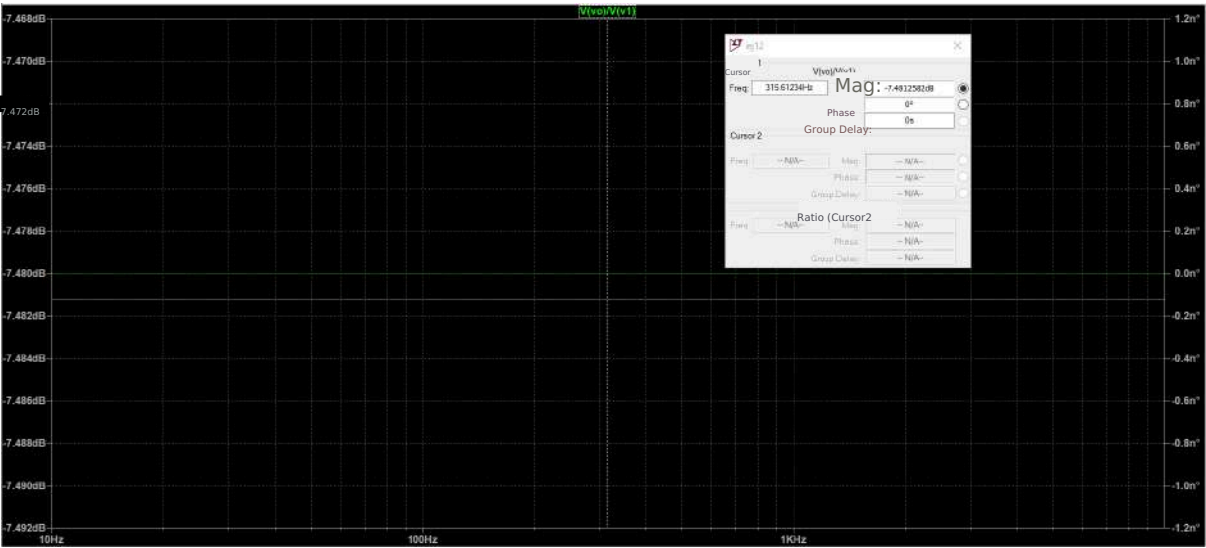


Gráfico 22

Circuit of exercise 12.



Graph 23

Frequency response of the voltage transfer of the circuit from exercise 12.

Exercise #13

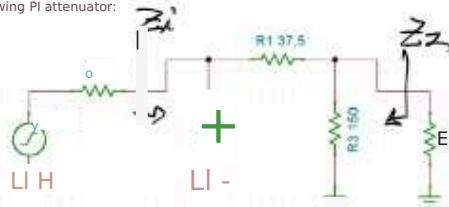
Exercise13 Sheet 1

Sunday, October 15, 2075

20:53

Exercise 1%.

Given the following PI attenuator:



$$S_{11} = \frac{Z_i - Z_{01}}{Z_i + Z_{01}}$$

$$S_{22} = \frac{Z_o - Z_{02}}{Z_o + Z_{02}}$$

- a) b) Empreendo Calculate the parameters parameters 5 of the \$, calculate quadripole, the power assuming measured at the attenuator and the power dissipated in cm
- 1) Compare the results obtained for the different values of Zo,
- €) How can you interpret the results of 611/622 and 521/512 obtained in both cases?

$$S_{a: +} \quad | \quad a \quad 7 \quad RE$$

Um aaron

o noldo: paa

$$S_R = \frac{V_1}{V_{g \text{ solida}}} \cdot 2$$

hos: Jean E

9) $Z_{01} = Z_{02} = Z_0$ entones - unant adaptada

$$Z_e = [(03/00) + 04]$$

$$S_{11} = \frac{Z_i - Z_{01}}{Z_i + Z_{01}}$$

$$S_{11} = \frac{S_a - 900}{E_{na} + 300}$$

$$\text{and } Z_i = ((150 \parallel 150) + 37.5) \parallel 150 = S_{en}$$

$$S_{11} = 0$$

$$S_{22} = \frac{Z_o - Z_{02}}{Z_o + Z_{02}}, \quad Z_o = ((09 \parallel 02) + 01) \parallel 03$$

$$S_{22} = \frac{500 - 500}{300 + 500}, \quad Z_o = ((500 \parallel 150) + 37.5) \parallel 150 = 500$$

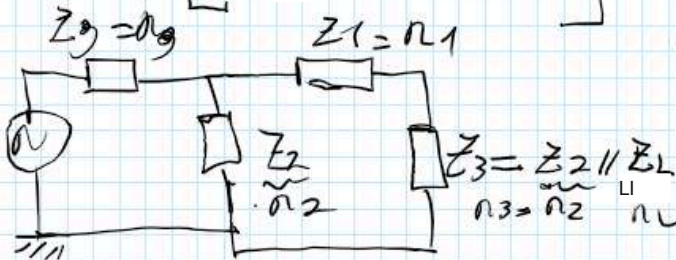
$$S_{22} = 0$$

Exercise13 Sheet

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$$S_{uz} = \frac{V_1}{V_{out}} \cdot 2 \sqrt{\frac{Z_{02}}{Z_{01}}}, \quad Z_{02} = Z_{01} \Rightarrow S_{uz} = \frac{V_1}{V_{out}} \cdot 2$$

$$[T] = \begin{bmatrix} 1 + Z \cdot Y & Z \\ Y & 1 \end{bmatrix} \quad \text{and} \quad \Rightarrow \quad \begin{array}{c} Z \\ \text{---} \\ Y \\ \text{---} \\ L \end{array}$$



$$[T] = \begin{bmatrix} 1 + \frac{1}{n_2} \cdot n_1 & n_1 \\ \frac{1}{n_2} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 + \frac{1}{n_2 \parallel n_L} \cdot n_1 & n_1 \\ \frac{1}{n_2 \parallel n_L} & 1 \end{bmatrix}$$

$$[T] = \begin{bmatrix} \left(1 + \frac{1}{n_2} \cdot n_1\right) \left(1 + \frac{1}{n_2 \parallel n_L} \cdot n_1\right) + n_1 \cdot \frac{1}{n_2 \parallel n_L} & \left(1 + \frac{1}{n_2} \cdot n_1\right) n_1 + n_1 \\ \frac{1}{n_2} \cdot \frac{1}{n_2 \parallel n_L} + \frac{1}{n_2 \parallel n_L} & \frac{1}{n_2} \cdot n_1 + 1 \end{bmatrix}$$

The

Ejercicio 13_Hoja_3

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$$A = \left(1 + \frac{l \cdot 0.9}{0.2}\right) \left(1 + \frac{l \cdot 0.9}{0.2 \parallel 1.1}\right) + \frac{l}{0.2 \parallel 1.1}$$

$$T = \left(1 + \frac{1}{150 \Omega} \cdot 37.5 \Omega\right) \left(1 + \frac{1}{150 \Omega \parallel 150 \Omega} \cdot 37.5 \Omega\right) + \frac{1}{150 \Omega \parallel 150 \Omega}$$

$$A = 1.33 \cdot 2 \cdot 1.33$$

$$A = 4 = \frac{1}{\frac{V_2}{V_0}} = \frac{V_0}{V_2}$$

$$\frac{V_2}{A} = \frac{V_0}{4} = \frac{1}{4} = 0.25$$

$$S_{21} = \frac{V_2}{V_0} \cdot 2 \cdot \sqrt{\frac{Z_{01}}{Z_{02}}}, \text{ con } Z_{01} = Z_{02}$$

$$S_{21} = \frac{V_2}{V_0} \cdot 2 \cdot 1 = 0.25 \cdot 2 = 0.5 = S_{21}$$

$$\text{Por simetría } \Rightarrow S_{21} = S_{12} = 0.5 \text{ y por simetría } S_{11} = S_{22}$$

$$b) G_T = \frac{P_{AL}}{P_{IN}} = \frac{P_L}{P_i} = |S_{21}|^2 = 0.5^2 = 0.25$$

$$G_T = \frac{P_{AL}}{P_A} = |S_{21}|^2 = 1$$

Ejercicio13_Hoja_4

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c) Con $Z_0 = TS = Z_{01} = Z_{02}$

$$S_{22} = \frac{Z_2 - Z_{02}}{Z_2 + Z_{02}}, \quad Z_2 = \frac{ES_{22} // R_2 + R_1}{AND}$$

$$S_{22} = \frac{55\Omega - 75\Omega}{55\Omega + 75\Omega}, \quad Z_2 = \frac{(75\Omega // 150\Omega) + 375\Omega}{150\Omega} = 35,26$$

$$S_{22} = -0,18$$

$$S_{11} = \frac{Z_1 - Z_{01}}{Z_1 + Z_{01}}, \quad Z_1 = \frac{(R_3 // R_L) + R_1}{R_2}, \quad Z_{01} = R_L = 75\Omega$$

$$S_{11} = \frac{55\Omega - 75\Omega}{55\Omega + 75\Omega}, \quad Z_1 = \frac{(150\Omega // 75\Omega) + 375\Omega}{150\Omega} = A$$

$$S_{11} = -0,18$$

$$A = \left(1 + \frac{l \cdot R_g}{R_2}\right) \left(\frac{1 + l \cdot R_g}{R_2 // R_L}\right) + R_g \frac{l}{R_2 // R_L}$$

$$A = \left(1 + \frac{l \cdot 75\Omega}{150\Omega}\right) \cdot \left(\frac{1 + l \cdot 375\Omega}{150\Omega // 75\Omega}\right) + 75\Omega \cdot \frac{l}{150\Omega // 75\Omega}$$

$$A = \frac{1}{2} \cdot 1,75 + 1,5$$

$$A = 2,375 \Rightarrow \frac{l}{A} = \frac{V_2}{V_1} = 0,42$$

$$S_{21} = \frac{V_2}{V_1} \cdot 2 \cdot \sqrt{\frac{Z_{01}}{Z_{02}}}, \quad \text{con } Z_{01} = Z_{02}$$

$$S_{21} = \frac{V_2}{V_1} \cdot 2 \cdot 1 = 0,42 \cdot 2 = 0,84 = S_{21}$$

Per

$$\text{for } S_{11} \Rightarrow S_{21} = S_{12} = 0,84$$

$$\text{y for } S_{22} \Rightarrow S_{11} = S_{22}$$

Ejercicio13_Hoja_5

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d) conclusiones

con $Z_{02} = Z_{01} = Z_0 = 50 \Omega$ It is $\Rightarrow S_{21} = 0,5$

con $Z_{02} = Z_{01} = Z_0 = 75 \Omega \Rightarrow S_{21} = 23H$

• Al saber $S_{21} \Rightarrow$ se observa más o menos $\Rightarrow$ hay mayor transmisión de potencia del "puerto 1 al 2".

• e) En $Z_0 = 50 \Omega \Rightarrow S_{11} = S_{22} = 0 \Rightarrow$ No hay potencia reflejada en puertos de entrada

$Z_0 = 75 \Omega \Rightarrow S_{11} = S_{22} = -0,5 \Rightarrow$ Hay algo de potencia reflejada en puertos de entrada y con desfase de 180° .

Exercise #14

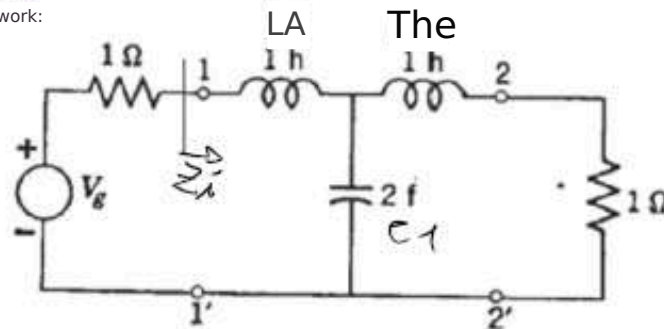
Ejercicio 14 Sheet 1

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18:49

Exercise 14.

For the following network:



A) Calculate the network parameters of the network.

B) Obtain analytically and plot $|S_{11}(v)|$ and $|S_{21}(v)|$. Draw conclusions about them in the stopband and attenuation band

5) $Z_g = Z_L = Z_e = f_s$

* Sus $\frac{Z_i - 1}{Z_i + 1} = s$ $Z_i = \left[(sL_2 + Z_g) \parallel \frac{1}{sC_1} \right] + sL_1$ $Z_i = Z_{g0}$ **OX A**

$Z = 7$ $\left[(sL_2 + 1) \parallel \frac{1}{sC_1} \right] + sL_1$

$Z_i = 9 \frac{sL_2 + 1}{3s^2L_2 + 1} 47 + \frac{sL_2 + 1 + sL_1(s^2C_1L_2 + 1)}{(s^2L_2 + 1)(s^2C_1L_2 + 1)}$

$Z_i = \frac{s^3C_1L_2L_1 + s^2C_1L_1 + s(L_2 + L_1)}{s^2L_2 + 1} 409$

$Z_i - Z_{01} = \frac{s^3C_1L_2L_1 + s^2C_1L_1 + s(L_2 + L_1) - (s^2L_2 + 1)}{s^2L_2 + 1} 9$ $[s^2L_2 + 1]$

$Z_i - Z_{01} = \frac{s^3C_1L_2L_1 + s^2C_1L_1 - (s^2L_2 + 1)}{s^2L_2 + 1} 9$

$Z_i - Z_{01} = \frac{s^3C_1L_2L_1 + s(L_2 + L_1 - C_1L_2)}{s^2L_2 + 1} 9$ $\frac{s^3 + 3.0}{s^2 + 1} = -4$ **try**

Exercise14 Sheet

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$$Z_i + Z_{01} = \frac{R_g^3 C_1 L_2 L_1 + R_g^2 C_1 L_1 + R_g (L_2 + L_1) + R_g + R_g (R_g C_1 L_2 + R_g C_1 L_1)}{R_g^2 C_1 L_2 + R_g C_1 L_1 + 1}$$

$$Z_i + Z_{01} = \frac{R_g^3 C_1 L_2 L_1 + R_g^2 (R_g C_1 L_1 + R_g C_1 L_2) + R_g (L_2 + L_1 + R_g^2 C_1) + R_g + R_g}{R_g^2 C_1 L_2 + R_g C_1 L_1 + 1}$$

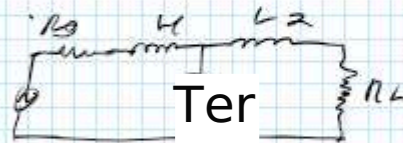
$$Z_i + Z_{01} = \frac{2R_g^3 + 4R_g^2 + 4R_g + 2}{2R_g^2 + 2R_g + 1}$$

$$S_{11} = \frac{2R_g^3}{2R_g^2 + 2R_g + 1} = \frac{2R_g^3}{2R_g^3 + 4R_g^2 + 4R_g + 2} = S_{11}$$

• Por simetría, se demuestra que $S_{11} = S_{22}$

$$S_{21} = \frac{V_2}{V_1} \cdot 2 \sqrt{\frac{Z_{01}}{Z_{02}}}, \quad Z_{01} = Z_{02} = R_g = R_L$$

$$S_{21} = \frac{V_2}{V_1} \cdot 2, \text{ Buses [T] de}$$



$$\text{1a celda en "L"} = \frac{R_g L_1}{C_1} \Rightarrow [T_1] = \begin{bmatrix} 1 + Z_1 Y_1 & Z_1 \\ Y_1 & 1 \end{bmatrix}$$

$$[T_1] = \begin{bmatrix} 1 + (R_g + j\omega L_1) \cdot j\omega C_1 & R_g + j\omega L_1 \\ j\omega C_1 & 1 \end{bmatrix}$$

$$\text{2da celda en "L"} = \frac{L_2}{R_L} \Rightarrow [T_2] = \begin{bmatrix} 1 + Z_2 Y_2 & Z_2 \\ Y_2 & 1 \end{bmatrix}$$

$$[T_2] = \begin{bmatrix} 1 + j\omega L_2 \cdot \frac{1}{R_L} & j\omega L_2 \\ \frac{1}{R_L} & 1 \end{bmatrix}$$

Exercise14 Sheet_3

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$$[T] = [T_1] \cdot [T_2]$$

$$[T] = \begin{bmatrix} (1 + (R_0 + \beta L_1) \beta C_1) \beta C_1 & R_0 + \beta L_1 \\ \beta C_1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 + \beta L_2 \frac{1}{R_L} & \beta L_2 \\ \frac{1}{R_L} & 1 \end{bmatrix}$$

(a)

$$[T] = \begin{bmatrix} (1 + (R_0 + \beta L_1) \beta C_1) \cdot (1 + \beta L_2 \frac{1}{R_L}) + (R_0 + \beta L_1) \cdot \frac{1}{R_L} & (1 + (R_0 + \beta L_1) \beta C_1) \beta L_2 + \beta L_1 \\ \beta C_1 (1 + \beta L_2 \frac{1}{R_L}) + \frac{1}{R_L} & \beta C_1 \beta L_2 + 1 \end{bmatrix}$$

$$A = (1 + (R_0 + \beta L_1) \beta C_1) \cdot (1 + \beta L_2 \frac{1}{R_L}) + (R_0 + \beta L_1) \cdot \frac{1}{R_L}$$

$$A = (1 + \beta C_1 R_0 + \beta^2 L_1 C_1) \cdot (1 + \beta \frac{L_2}{R_L}) + R_0 \frac{1}{R_L} + \beta \frac{L_1}{R_L}$$

$$A = 1 + \beta \frac{L_2}{R_L} + \beta C_1 R_0 + \beta^2 C_1 R_0 \frac{L_2}{R_L} + \beta^2 L_1 C_1 \cdot 1 + \beta^3 L_1 C_1 \frac{L_2}{R_L} + R_0 \frac{1}{R_L} + \beta \frac{L_1}{R_L}$$

$$A = \beta^3 L_1 C_1 \frac{L_2}{R_L} + \beta^2 (C_1 R_0 \frac{L_2}{R_L} + L_1 \cdot C_1) + \beta (\frac{L_2}{R_L} + C_1 R_0 + \frac{L_1}{R_L}) + 1 + R_0 \frac{1}{R_L}$$

$$A = 2\beta^3 + 4\beta^2 + 4\beta + 2 = (\beta^3 + 2\beta^2 + 2\beta + 1) \cdot 2 = A$$

$$\frac{L}{A} = \frac{V_2}{V_0} \Rightarrow$$

$$\beta_{21} = \frac{V_2}{V_0} \cdot 2 = \frac{1}{(\beta^3 + 2\beta^2 + 2\beta + 1) \cdot 2} \cdot 2 = \frac{1}{\beta^3 + 2\beta^2 + 2\beta + 1} = \beta_{21}$$

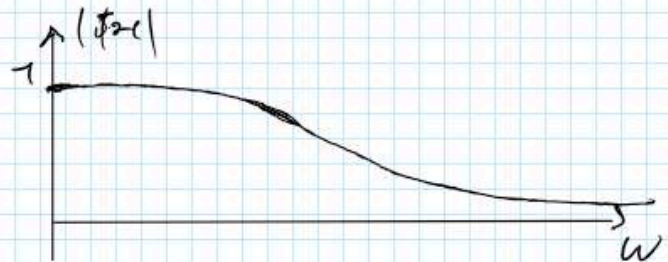
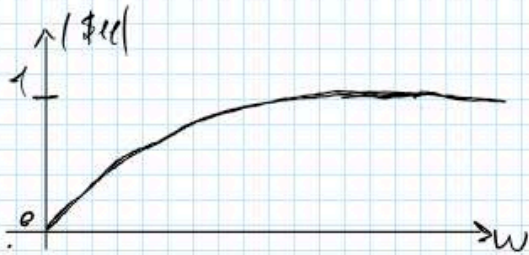
• Por ser circuito pasivo, se demuestra que $\beta_{21} = \beta_{12}$

Ejercicio14_H^{Sheet}a_4

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$$\frac{2\delta^3}{2\delta^3 + 4\delta^2 + 4\delta + 2} = \delta_{11}$$

$$\frac{1}{\delta^3 + 2\delta^2 + 2\delta + 1} = \delta_{21}$$



- Del circuito, se obtiene una red o poro largo.
- En la banda pasabaja vale como poro largo $w \rightarrow 0$ y $|\delta_{21}| \approx 1$ poro largo
- Poro largo permeable, hay match perfecto, todo lo que se envía se transmite en términos de potencia